MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

ENCODER, PARITY, DIGITAL, KEY CODE MHK

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic parity encoder module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 2.093 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be MHK.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/77-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

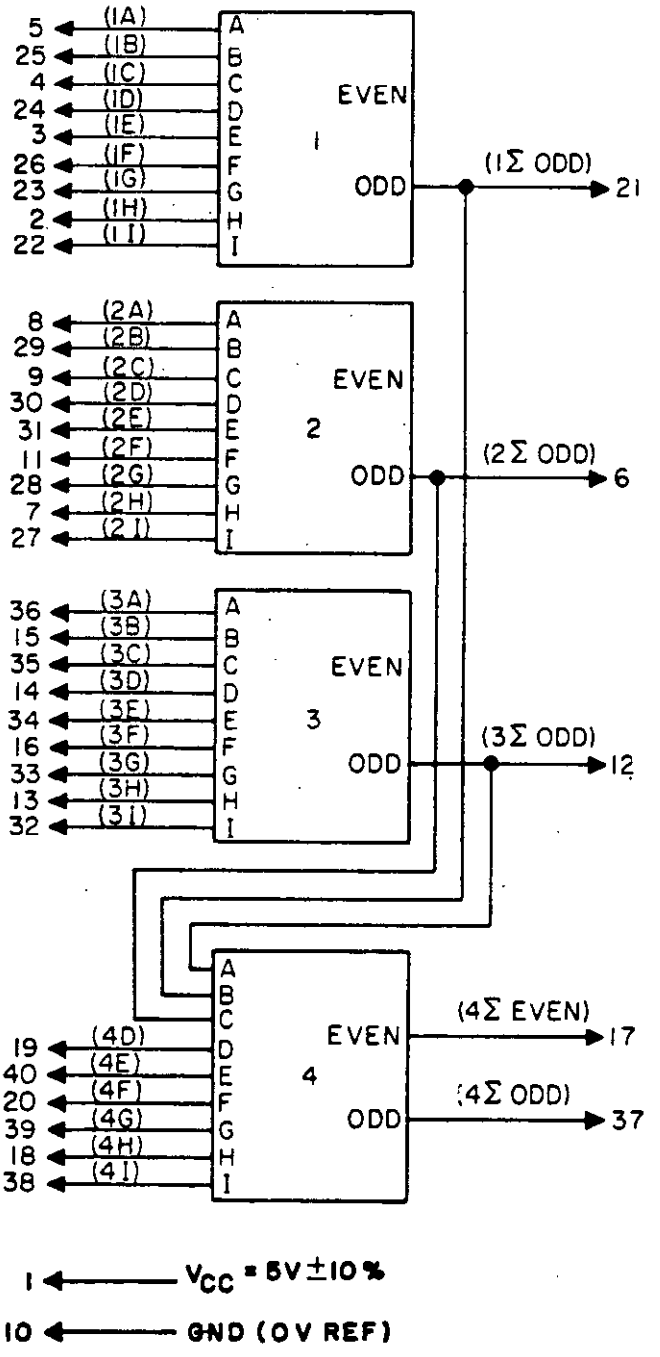
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. 1A-1I, 2A-2I, 3A-3I, 4D-4I = Data inputs.
2. 1Σ odd, 2Σ odd, 3Σ odd, 4Σ odd = Odd parity outputs.
3. 4Σ even = Even parity output.

* FIGURE 1. MHK module function diagram, parity encoder.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B5	V_{OUT} (low loaded)	IV	S	0.01	1.00	0.01	1.00	0.01	1.00	0.01	1.00	V
C1-C33	Input breakdown current	V	S	0	1	0	1	0	1	0	1	mA
D1-D5	Output leakage current	VI	X	0	1	0	1	0	1	0	1	mA
E1-E5	Output short circuit current	VII	X	-40	-100	-40	-100	-40	-100	-40	-100	mA
F1-F19	Functional	VIII	X	Test vectors must product output vectors as specified in table VIII.								
G1-G33	Input clamp diode voltage	IX	X	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
H1-H5	Output voltage, low level	X	X	0	500	0	500	0	500	0	500	mV
J1-J5	Output voltage, high level	XI	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
K1-K33	Input current, high level	XII	X	0	50	0	50	0	50	0	50	µA
L1-L33	Input current, low level	XIII	X	-0.1	-2.2	-0.1	-2.2	-0.1	-2.2	-0.1	-2.2	mA
M	Power supply current	XIV	X	100	396	100	396	100	396	100	396	mA
N1-N5	Transition time (t_{TLH}), C_L = 50 pF	XV	S	2	15	2	15	2	18	2	18	ns
N6-N10	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	2	27	2	27	2	30	2	30	ns
P1-P5	Transition time (t_{THL}), C_L = 50 pF	XV	S	2	6	2	6	2	8	2	8	ns
P6-P10	Transition time (t_{THL}), C_L = 150 pF	XV	Q	2	11	2	11	2	13	2	13	ns
Q1-Q21	Propagation delay time (t_{PLH}), C_L = 50 pF	XVI	S	5.0	27.5	5.0	27.5	5.0	29	5	29	ns
Q22-Q27	Propagation delay time (t_{PLH}), C_L = 50 pF	XVI	S	5	55	5	55	5	58	5	58	ns

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q28-Q48	Propagation delay time (t_{PLH}), C_L = 150 pF	XVI	Q	5	32	5	32	5	35	5	35	ns
Q49-Q54	Propagation delay time (t_{PLH}), C_L = 150 pF	XVI	Q	5	64	5	64	5	70	5	70	ns
R1-R23	Propagation delay time (t_{PHL}), C_L = 50 pF	XVI	S	5	24	5	24	5.0	25.5	5.0	25.5	ns
R24-R29	Propagation delay time (t_{PHL}), C_L = 50 pF	XVI	S	5.0	51.5	5.0	51.5	5.0	54.5	5.0	54.5	ns
R30-R52	Propagation delay time (t_{PHL}), C_L = 150 pF	XVI	Q	5	28	5	28	5	31	5	31	ns
R53-R58	Propagation delay time (t_{PHL}), C_L = 150 pF	XVI	Q	5	60	5	60	5	66	5	66	ns
S1-S3	Output crosstalk, low level	XVII	Q	0	500	0	500	0	500	0	500	mV
T1-T3	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	mV
U	Power supply transient amplitude	XVIII	Q	0	400	0	400	0	400	0	400	mV
V	Power supply filter capacitance	XIX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
W1-W33	Input capacitance	XX	Q	5	15	5	15	5	15	5	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition V, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through V shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/77-1 supersedes part number M28787/77. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation	2,178 mW (outputs not loaded)
Supply voltage	5 V dc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2 V minimum, V_{CC} maximum

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Input current	
Low level (V_{IL} = 0 V)	-2.2 mA maximum
Low level (V_{IL} = 0.5 V)	-2 mA maximum
Input current	
High level (V_{IH} = 2.5 V)	50 µA maximum
High level (V_{IH} = 5.5 V)	1 mA maximum
Output voltage,	
1Σ odd, 2Σ odd, 3Σ odd	
Low level (I_{OL} = 18 mA)	0.5 V maximum
High level (I_{OH} = -950 µA)	2.5 V minimum
Output voltage, 4Σ odd, 4Σ even	
Low level (I_{OL} = 20 mA)	0.5 V maximum
High level (I_{OH} = -1 mA)	2.5 V minimum
Output drive current,	
1Σ odd, 2Σ odd, 3Σ odd,	
maximum capability	
Low level (V_{OL} = 0.5 V)	18 mA
High level (V_{OH} = -2.5 V)	-950 µA

* 6.5 Module function. This module contains four 9-bit parity encoders interconnected as one 33-bit parity encoder.

6.5.1 Input to output relationship. Each parity encoder has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/ - 3/

Number of inputs A thru I** that are high	Outputs	
	Σ Even	Σ Odd
0, 2, 4, 6, 8	H	L
1, 3, 5, 7, 9	L	H

1/ H = High-voltage level.
2/ L = Low-voltage level
3/ ** = Parity encoder 4 obtains inputs A, B, C from 3Σ odd, 1Σ odd, and 2Σ odd respectively.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

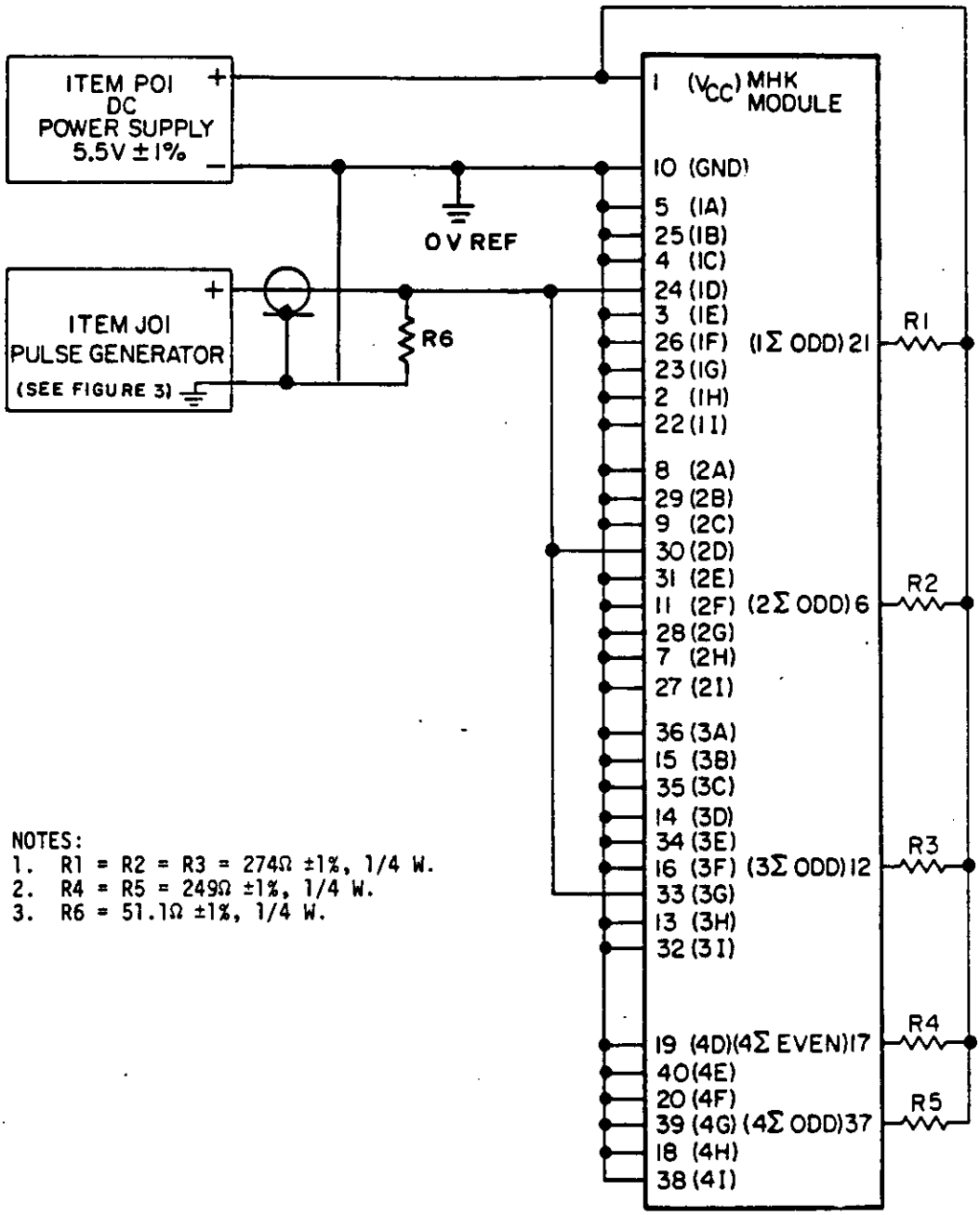
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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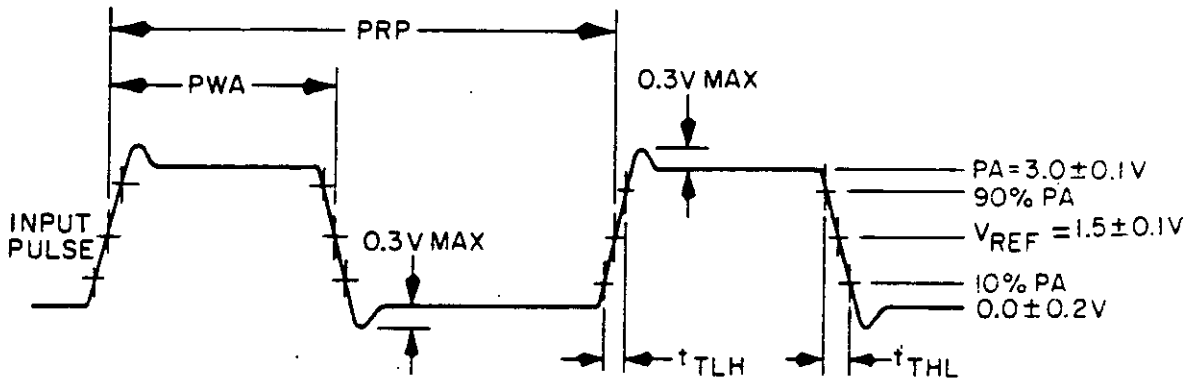
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NOTES:
1. R1 = R2 = R3 = 274Ω ±1%, 1/4 W.
2. R4 = R5 = 249Ω ±1%, 1/4 W.
3. R6 = 51.1Ω ±1%, 1/4 W.

FIGURE 2. Operating temperature and life test circuit.

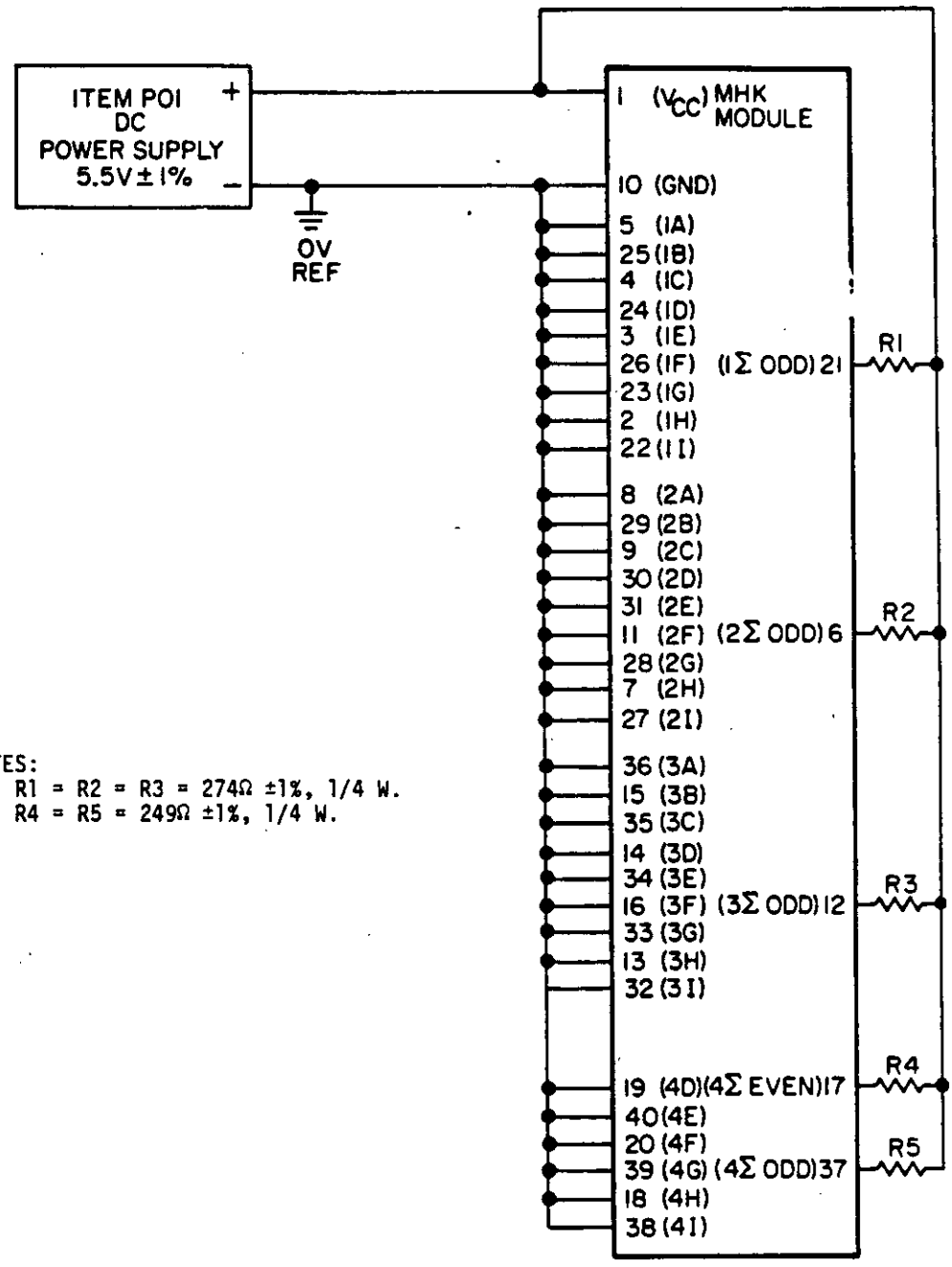
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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} \leq 10$ ns.
PRP = 1,000 ±200 ns.
PWA = 500 ±100 ns.

FIGURE 3. Operating temperature and life test circuit waveform.

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NOTES:
1. R1 = R2 = R3 = 274Ω ±1%, 1/4 W.
2. R4 = R5 = 249Ω ±1%, 1/4 W.

FIGURE 4. Component temperature test circuit.

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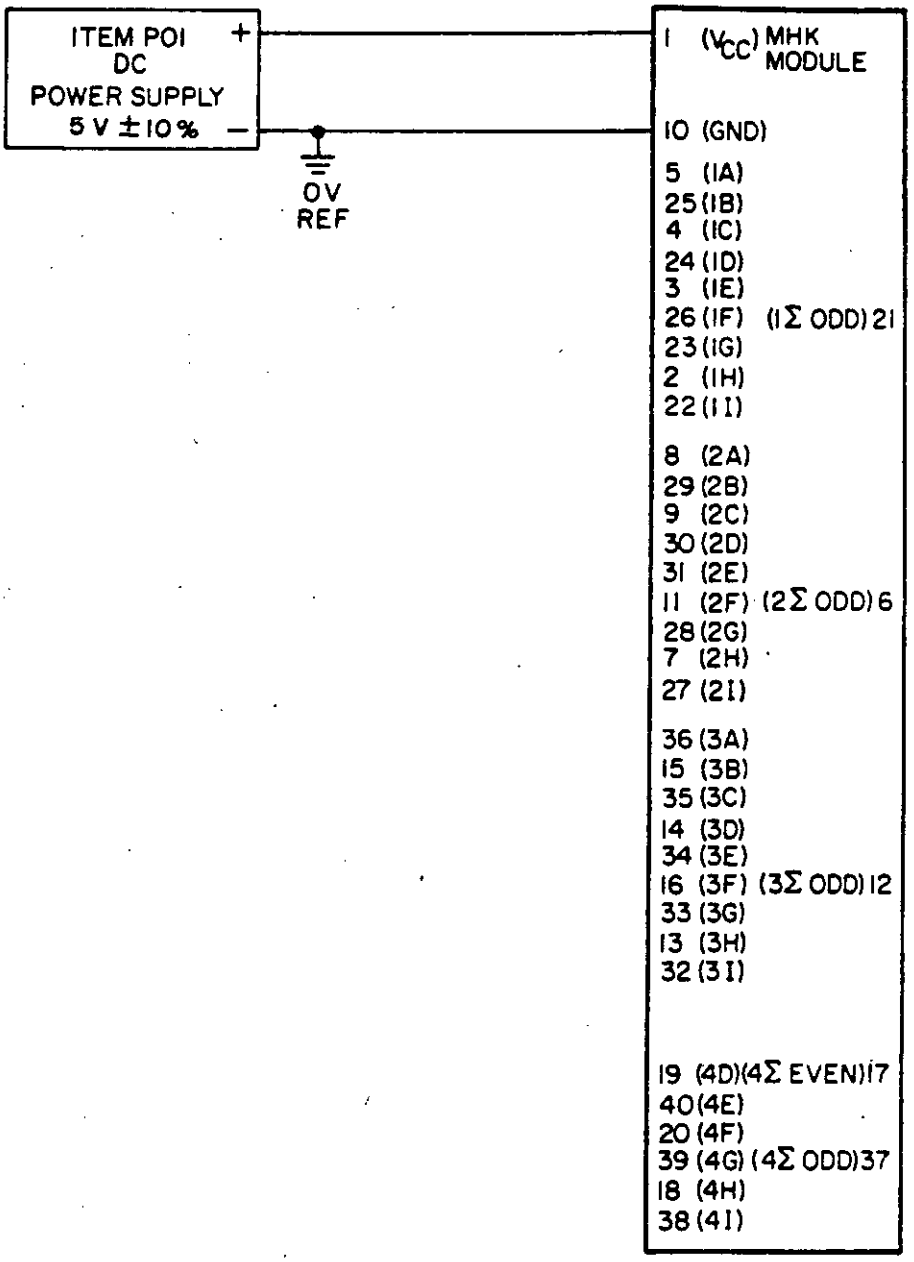


FIGURE 5. Vibration test circuit.

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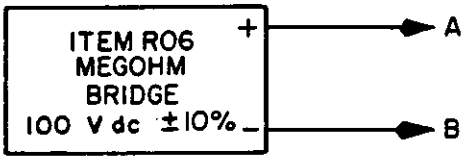
TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	All pins	Frame	10	∞	MΩ	6

TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1Σ odd V_{OUT} (low loaded)	5	21	0.01	1.00	V	7
B2	2Σ odd "	8	6	"	"	"	"
B3	3Σ odd "	36	12	"	"	"	"
B4	4Σ odd "	19	37	"	"	"	"
B5	4Σ even "	Open	17	"	"	"	"

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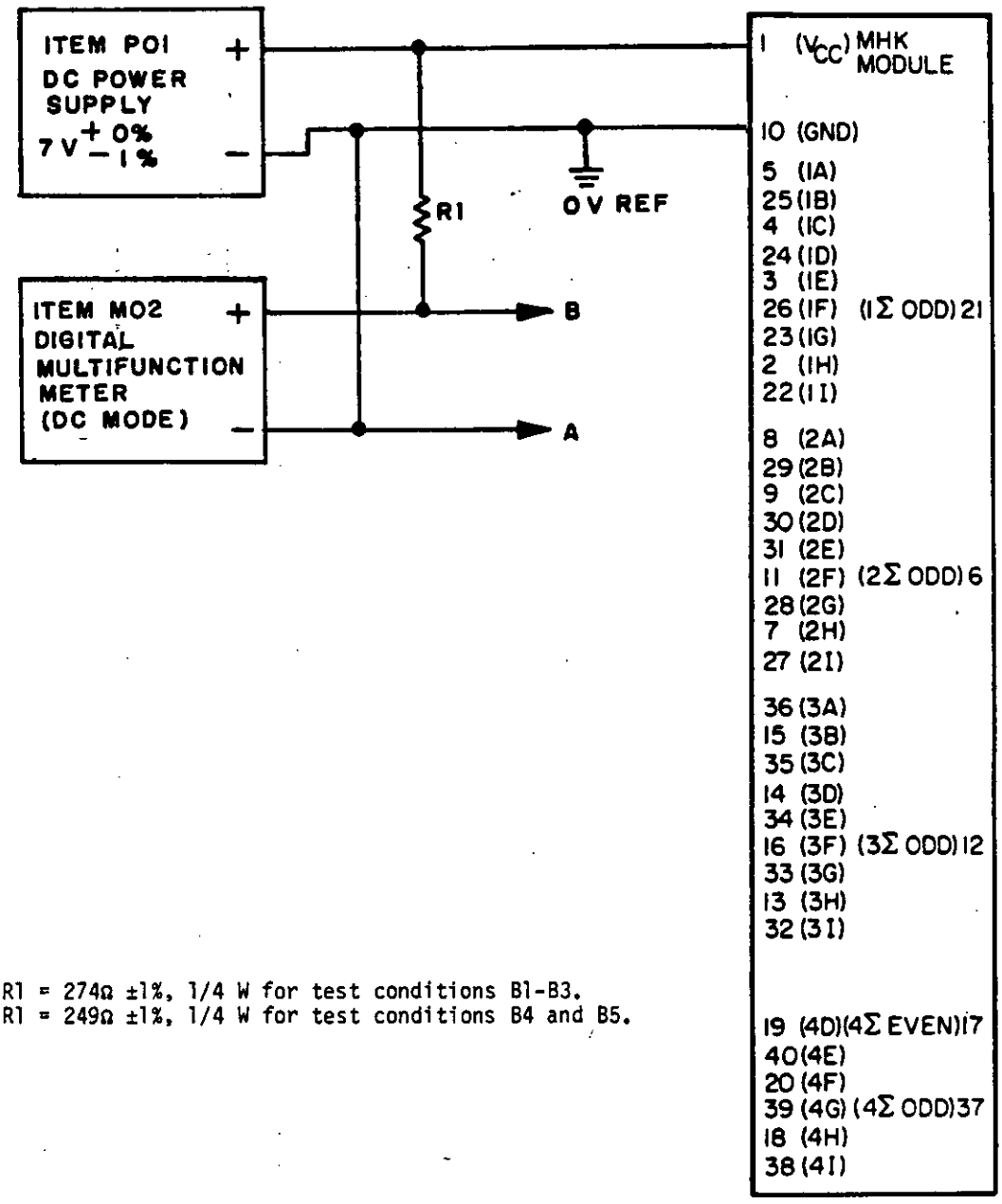
When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

1	(V_{CC}) MHK MODULE	
10	(GND)	
5	(1A)	
25	(1B)	
4	(1C)	
24	(1D)	
3	(1E)	
26	(1F)	(1Σ ODD) 21
23	(1G)	
2	(1H)	
22	(1I)	
8	(2A)	
29	(2B)	
9	(2C)	
30	(2D)	
31	(2E)	
11	(2F)	(2Σ ODD) 6
28	(2G)	
7	(2H)	
27	(2I)	
36	(3A)	
15	(3B)	
35	(3C)	
14	(3D)	
34	(3E)	
16	(3F)	(3Σ ODD) 12
33	(3G)	
13	(3H)	
32	(3I)	
19	(4D)	(4Σ EVEN) 17
40	(4E)	
20	(4F)	
39	(4G)	(4Σ ODD) 37
18	(4H)	
38	(4I)	

FIGURE 6. Isolation test circuit for table III.

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NOTE: R1 = 274Ω ±1%, 1/4 W for test conditions B1-B3.
R1 = 249Ω ±1%, 1/4 W for test conditions B4 and B5.

* FIGURE 7. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1A Input breakdown current	2,3,4,7,8,9,11,13,14,15,16, 18,19,20,22,23,24,25,26,27, 28,29,30,31,32,33,34,35,36, 38,39,40	5	0	1	mA	8
C2	1B "	2,3,4,5,7,8,9,11,13,14,15, 16,18,19,20,22,23,24,26,27, 28,29,30,31,32,33,34,35,36, 38,39,40	25	"	"	"	"
C3	1C "	2,3,5,7,8,9,11,13,14,15, 16,18,19,20,22,23,24,25,26, 27,28,29,30,31,32,33,34,35, 36,38,39,40	4	"	"	"	"
C4	1D "	2,3,4,5,7,8,9,11,13,14,15, 16,18,19,20,22,23,25,26, 27,28,29,30,31,32,33,34,35, 36,38,39,40	24	"	"	"	"
C5	1E "	2,4,5,7,8,9,11,13,14,15, 16,18,19,20,22,23,24,25,26, 27,28,29,30,31,32,33,34,35, 36,38,39,40	3	"	"	"	"
C6	1F "	2,3,4,5,7,8,9,11,13,14,15, 16,18,19,20,22,23,24,25, 27,28,29,30,31,32,33,34,35, 36,38,39,40	26	"	"	"	"
C7	1G "	2,3,4,5,7,8,9,11,13,14,15, 16,18,19,20,22,24,25,26, 27,28,29,30,31,32,33,34,35, 36,38,39,40	23	"	"	"	"
C8	1H "	3,4,5,7,8,9,11,13,14,15, 16,18,19,20,22,23,24,25,26, 27,28,29,30,31,32,33,34,35, 36,38,39,40	2	"	"	"	"
C9	1I "	2,3,4,5,7,8,9,11,13,14,15, 16,18,19,20,23,24,25,26, 27,28,29,30,31,32,33,34,35, 36,38,39,40	22	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C10	2A	"	2,3,4,5,7,9,11,13,14,15, 16,18,19,20,22,23,24,25,26, 27,28,29,30,31,32,33,34,35, 36,38,39,40	8	0	1	mA	8
C11	2B	"	2,3,4,5,7,8,9,11,13,14,15, 16,18,19,20,22,23,24,25,26, 27,28,30,31,32,33,34,35, 36,38,39,40	29	"	"	"	"
C12	2C	"	2,3,4,5,7,8,11,13,14,15, 16,18,19,20,22,23,24,25,26, 27,28,29,30,31,32,33,34,35, 36,38,39,40	9	"	"	"	"
C13	2D	"	2,3,4,5,7,8,9,11,13,14,15, 16,18,19,20,22,23,24,25,26, 27,28,29,31,32,33,34,35, 36,38,39,40	30	"	"	"	"
C14	2E	"	2,3,4,5,7,8,9,11,13,14,15, 16,18,19,20,22,23,24,25,26, 27,28,29,30,32,33,34,35, 36,38,39,40	31	"	"	"	"
C15	2F	"	2,3,4,5,7,8,9,13,14,15, 16,18,19,20,22,23,24,25,26, 27,28,29,30,31,32,33,34,35, 36,38,39,40	11	"	"	"	"
C16	2G	"	2,3,4,5,7,8,9,11,13,14,15, 16,18,19,20,22,23,24,25,26, 27,29,30,31,32,33,34,35, 36,38,39,40	28	"	"	"	"
C17	2H	"	2,3,4,5,8,9,11,13,14,15, 16,18,19,20,22,23,24,25,26, 27,28,29,30,31,32,33,34,35, 36,38,39,40	7	"	"	"	"
C18	2I	"	2,3,4,5,7,8,9,11,13,14,15, 16,18,19,20,22,23,24,25,26, 28,29,30,31,32,33,34,35, 36,38,39,40	27	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

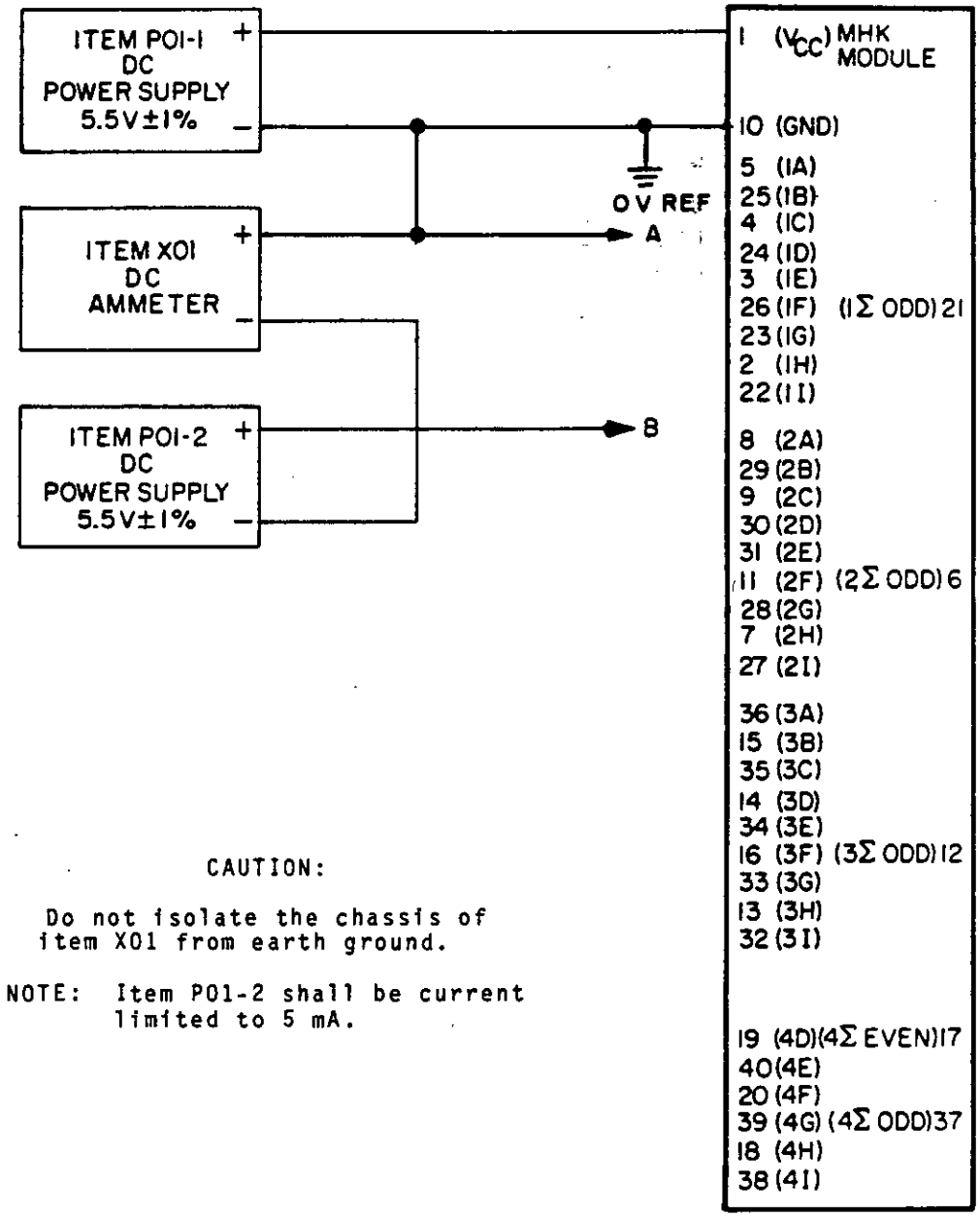
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C19	3A	Input breakdown current	2,3,4,5,7,8,9,11,13,14,15, 16,18,19,20,22,23,24,25,26, 27,28,29,30,31,32,33,34,35, 38,39,40	36	0	1	mA	8
C20	3B	"	2,3,4,5,7,8,9,11,13,14, 16,18,19,20,22,23,24,25,26, 27,28,29,30,31,32,33,34,35, 36,38,39,40	15	"	"	"	"
C21	3C	"	2,3,4,5,7,8,9,11,13,14,15, 16,18,19,20,22,23,24,25,26, 27,28,29,30,31,32,33,34, 36,38,39,40	35	"	"	"	"
C22	3D	"	2,3,4,5,7,8,9,11,13,15, 16,18,19,20,22,23,24,25,26, 27,28,29,30,31,32,33,34,35, 36,38,39,40	14	"	"	"	"
C23	3F	"	2,3,4,5,7,8,9,11,13,14,15, 16,18,19,20,22,23,24,25,26, 27,28,29,30,31,32,33,35, 36,38,39,40	34	"	"	"	"
C24	3F	"	2,3,4,5,7,8,9,11,13,14,15, 18,19,20,22,23,24,25,26, 27,28,29,30,31,32,33,34,35, 36,38,39,40	16	"	"	"	"
C25	3G	"	2,3,4,5,7,8,9,11,13,14,15, 16,18,19,20,22,23,24,25,26, 27,28,29,30,31,32,34,35, 36,38,39,40	33	"	"	"	"
C26	3H	"	2,3,4,5,7,8,9,11,14,15, 16,18,19,20,22,23,24,25,26, 27,28,29,30,31,32,33,34,35, 36,38,39,40	13	"	"	"	"
C27	3I	"	2,3,4,5,7,8,9,11,13,14,15, 16,18,19,20,22,23,24,25,26, 27,28,29,30,31,33,34,35, 36,38,39,40	32	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C28	4D Input breakdown current	2,3,4,5,7,8,9,11,13,14,15,16,18,20,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,38,39,40	19	0	1	mA	8
C29	4E "	2,3,4,5,7,8,9,11,13,14,15,16,18,19,20,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,38,39,	40	"	"	"	"
C30	4F "	2,3,4,5,7,8,9,11,13,14,15,16,18,19,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,38,39,40	20	"	"	"	"
C31	4G "	2,3,4,5,7,8,9,11,13,14,15,16,18,19,20,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,38,40	39	"	"	"	"
C32	4H "	2,3,4,5,7,8,9,11,13,14,15,16,19,20,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,38,39,40	18	"	"	"	"
C33	4I "	2,3,4,5,7,8,9,11,13,14,15,16,18,19,20,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,39,40	38	"	"	"	"

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

NOTE: Item PO1-2 shall be current limited to 5 mA.

FIGURE 8. Input breakdown current test circuit for table V.

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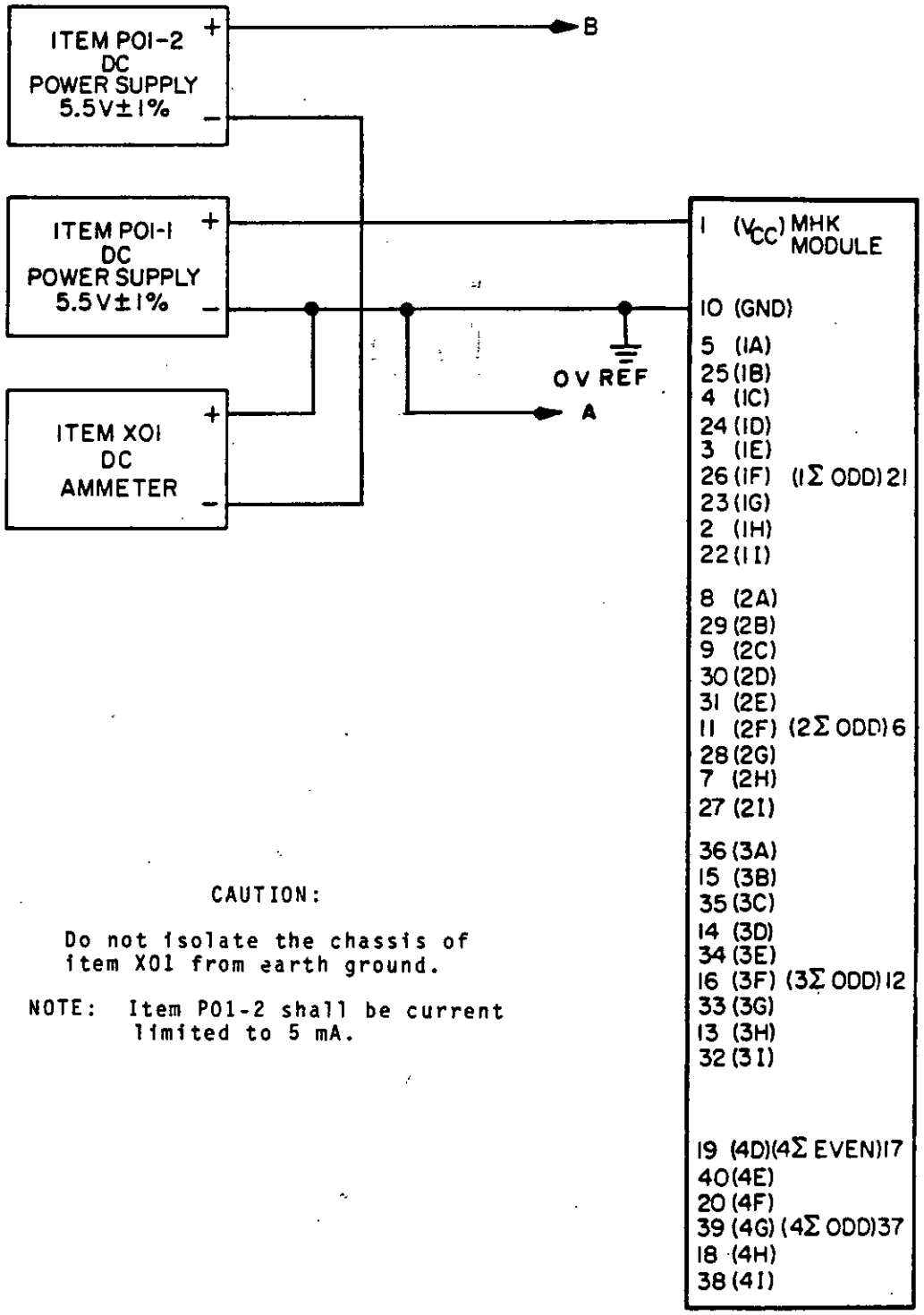
TABLE VI. Output leakage current test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
D1	1Σ odd	Output leakage current	Open	21	0	1	mA	9
D2	2Σ odd	"	"	6	"	"	"	"
D3	3Σ odd	"	"	12	"	"	"	"
D4	4Σ odd	"	"	37	"	"	"	"
D5	4Σ even	"	19	17	"	"	"	"

TABLE VII. Output short circuit current test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
E1	1Σ odd	Output short circuit current	8,19,36	21	-40	-100	mA	10
E2	2Σ odd		5,19,36	6	"	"	"	"
E3	3Σ odd		5,8,19	12	"	"	"	"
E4	4Σ odd		5,8,36	37	"	"	"	"
E5	5Σ even		5,8,19,36	17	"	"	"	"

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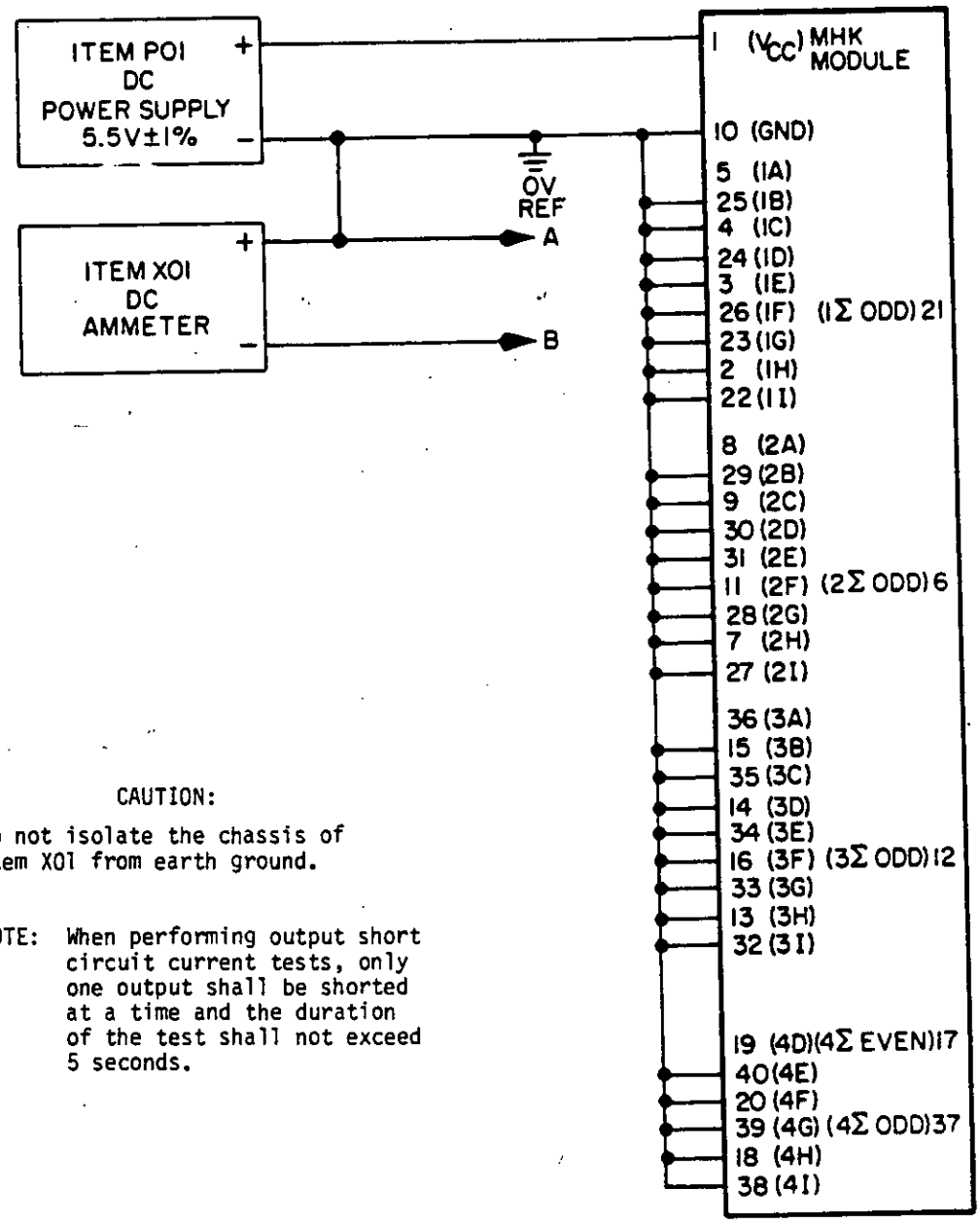
CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

NOTE: Item PO1-2 shall be current limited to 5 mA.

FIGURE 9. Output leakage current test circuit for table VI.

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTE: When performing output short circuit current tests, only one output shall be shorted at a time and the duration of the test shall not exceed 5 seconds.

FIGURE 10. Output short circuit current test circuit for table VII.

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TABLE VIII. Functional test.

Test condition	Parameter	Connect			Output logic states					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Σ odd	2Σ odd	3Σ odd	4Σ odd	4Σ even	Min	Max		
F1	Functional	4,8,11,13,15,16, 23,24,25,29,30, 32,34,36	2,3,5,7,9,14,18, 19,20,22,26,27, 28,31,33,35,38, 39,40	2/ 3/	L	L	L	L	H		1/		11
F2	"	2,3,8,9,15,18, 19,22,24,28,30, 31,32,33,35,39	4,5,7,11,13,14, 16,20,23,25,26, 27,29,34,36,38, 40	"	L	H	L	L	H		"		"
F3	"	2,5,11,13,15,22, 24,25,27,28,30, 31,32,34,38,39	3,4,7,8,9,14,16, 18,19,20,23,26, 29,33,35,36,40	"	H	H	L	L	H		"		"
F4	"	4,5,7,8,11,14, 15,16,18,22,23, 24,25,26,27,30, 32,33,34	2,3,9,13,19,20, 28,29,31,35,36, 38,39,40	"	H	H	L	H	L		"		"
F5	"	3,4,5,9,11,14, 15,19,23,24,27, 28,30,31,35	2,7,8,13,16,18, 20,22,25,26,29, 32,33,34,36,38, 39,40	"	H	L	H	H	L		"		"
F6	"	5,7,8,9,11,13, 14,15,16,18,20, 22,24,25,27,31, 34,35,38,39,40	2,3,4,19,23,26, 28,29,30,32,33, 36	"	L	L	L	H	L		"		"
F7	"	3,5,7,8,9,13,14, 15,20,25,26,27, 28,29,33,35,38	2,4,11,16,18,19, 22,23,24,30,31, 32,34,36,39,40	"	L	L	H	H	L		"		"
F8	"	2,3,5,7,9,13,15, 16,18,23,24,25, 26,27,28,29,30, 34,35,36,38	4,8,11,14,19,20, 22,31,32,33,39, 40	"	H	L	L	H	L		"		"
F9	"	3,5,8,11,13,14, 18,20,23,24,29, 30,35,36,38,39	2,4,7,9,15,16, 19,22,25,26,27, 28,31,32,33,34, 40	"	L	L	L	L	H		"		"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect			Output logic states					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Σ odd	2Σ odd	3Σ odd	4Σ odd	4Σ even	Min	Max		
F10	Functional	2,8,9,11,16,18, 19,20,22,23,24, 25,26,28,32,38, 40	3,4,5,7,13,14, 15,27,29,30,31, 33,34,35,36,39	2/ 3/	L	L	L	H	L		1/		11
F11	"	2,8,9,11,15,16, 22,23,25,27,29, 30,33,34,35,39	3,4,5,7,13,14, 18,19,20,24,26, 28,31,32,36,38, 40	"	L	L	H	L	H		"		"
F12	"	4,5,9,13,14,15, 16,19,20,26,27, 28,31,35,38	2,3,7,8,11,18, 22,23,24,25,29, 30,32,33,34,36, 39,40	"	H	L	H	H	L		"		"
F13	"	4,5,7,8,13,14, 18,19,24,32,33, 34,36,38,39	2,3,9,11,15,16, 20,22,23,25,26, 27,28,29,30,31, 35,40	"	H	L	L	H	L		"		"
F14	"	4,5,7,8,9,14,15, 16,20,22,28,30, 33,39	2,3,11,13,18,19, 23,24,25,26,27, 29,31,32,34,35, 36,38,40	"	H	H	L	L	H		"		"
F15	"	2,3,11,13,14,16, 18,20,23,25,27, 28,29,30,31,32, 36,38	4,5,7,8,9,15,19, 22,24,26,33,34, 35,39,40	"	L	L	H	L	H		"		"
F16	"	4,5,7,16,18,22, 23,24,25,29,33, 35	2,3,8,9,11,13, 14,15,19,20,26, 27,28,30,31,32, 34,36,38,39,40	"	L	L	H	L	H		"		"
F17	"	2,3,4,7,11,14, 15,19,27,28,29, 31,32,35,36,39, 40	5,8,9,13,16,18, 20,22,23,24,25, 26,30,33,34,38	"	H	L	H	H	L		"		"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued.

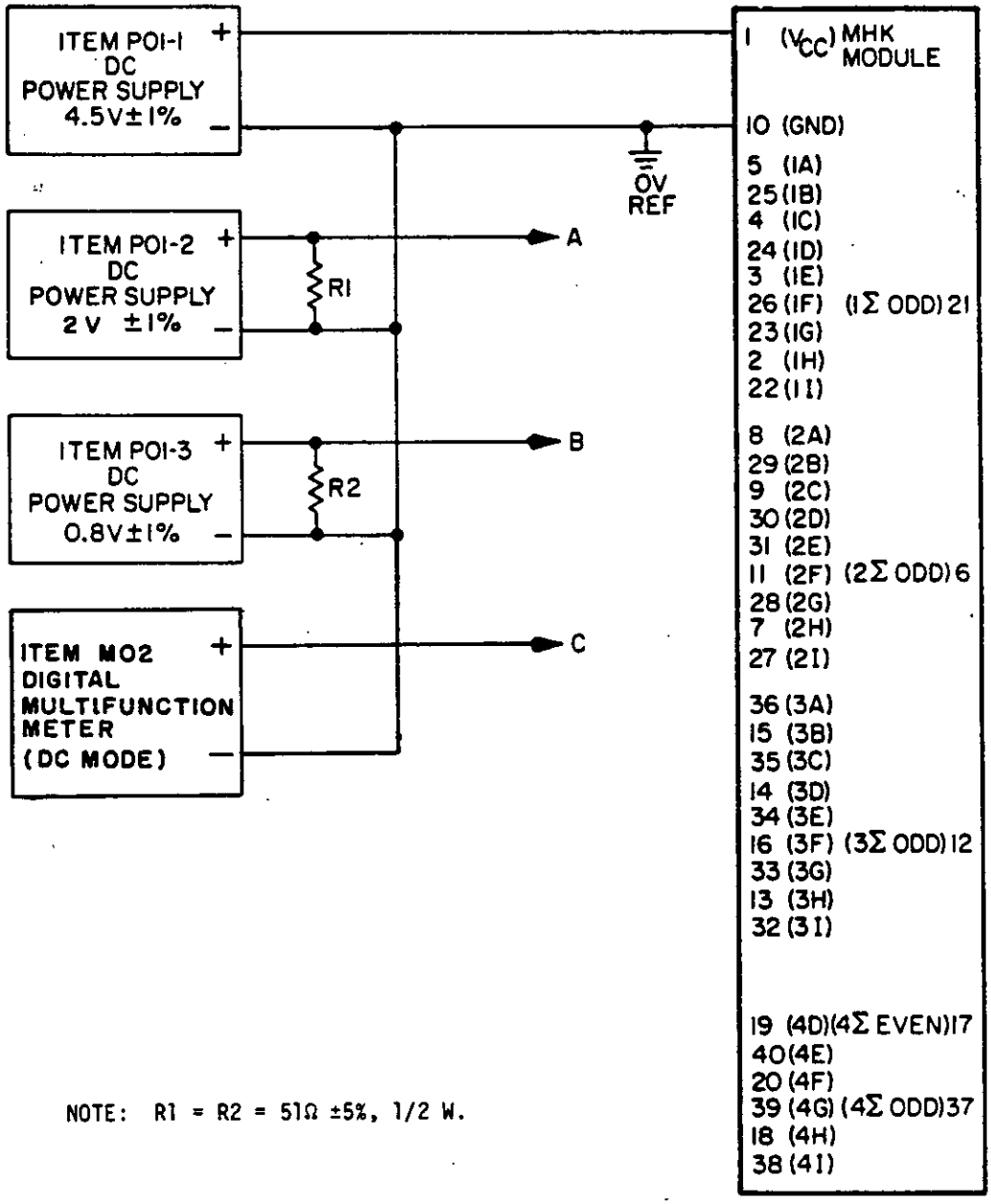
Test condition	Parameter	Connect			Output logic states					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1Σ odd	2Σ odd	3Σ odd	4Σ odd	4Σ even	Min	Max		
F18	Functional	5,8,14,18,19,20, 23,24,28,30,33, 36,38	2,3,4,7,9,11,13, 15,16,22,25,26, 27,29,31,32,34, 35,39,40	2/ 3/	H	H	H	H	L		1/		11
F19	Functional	3,5,8,14,15,22, 26,28,30,33,39, 40	2,4,7,9,11,13, 16,18,19,20,23, 24,25,27,29,31, 32,34,35,36,38	2/ 3/	L	H	H	L	H		1/		11

1/ Output logic states:
L = Low level = 0.0 to 0.5 V.
H = High level = 2.5 to 4.5 V.

2/ For test conditions F1-F19, verification of the output logic states in footnote 1 is obtained by connecting item M02 (point C) to each output pin in succession after input conditions have been accomplished.

3/ 1Σ odd, 2Σ odd, 3Σ odd, 4Σ odd, and 4Σ even output logic states are measured at output pins 21, 6, 12, 37, and 17, respectively.

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NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

* FIGURE 11. Functional test circuit for table VIII.

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TABLE IX. Input clamp diode voltage test.

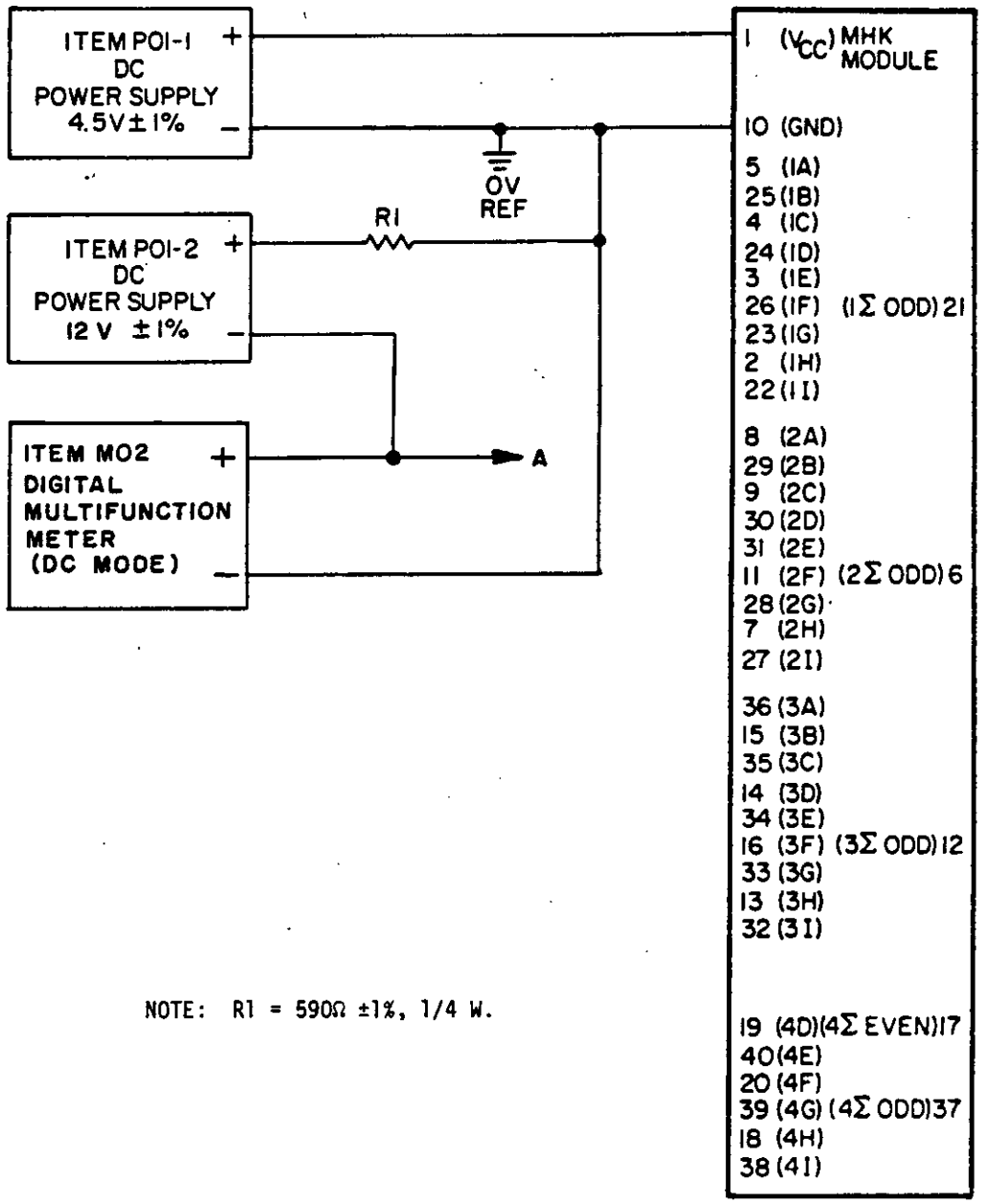
Test condition		Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
G1	1A	Input clamp diode voltage	5	-0.4	-1.2	V	12
G2	1B	"	25	"	"	"	"
G3	1C	"	4	"	"	"	"
G4	1D	"	24	"	"	"	"
G5	1E	"	3	"	"	"	"
G6	1F	"	26	"	"	"	"
G7	1G	"	23	"	"	"	"
G8	1H	"	2	"	"	"	"
G9	1I	"	22	"	"	"	"
G10	2A	"	8	"	"	"	"
G11	2B	"	29	"	"	"	"
G12	2C	"	9	"	"	"	"
G13	2D	"	30	"	"	"	"
G14	2E	"	31	"	"	"	"
G15	2F	"	11	"	"	"	"
G16	2G	"	28	"	"	"	"
G17	2H	"	7	"	"	"	"
G18	2I	"	27	"	"	"	"
G19	3A	"	36	"	"	"	"
G20	3B	"	15	"	"	"	"
G21	3C	"	35	"	"	"	"

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TABLE IX. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
G22	3D	Input clamp diode voltage	14	-0.4	-1.2	V	12
G23	3E	"	34	"	"	"	"
G24	3F	"	16	"	"	"	"
G25	3G	"	33	"	"	"	"
G26	3H	"	13	"	"	"	"
G27	3I	"	32	"	"	"	"
G28	4D	"	19	"	"	"	"
G29	4E	"	40	"	"	"	"
G30	4F	"	20	"	"	"	"
G31	4G	"	39	"	"	"	"
G32	4H	"	18	"	"	"	"
G33	4I	"	38	"	"	"	"

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NOTE: R1 = 590Ω ±1%, 1/4 W.

* FIGURE 12. Input clamp diode voltage test circuit for table IX.

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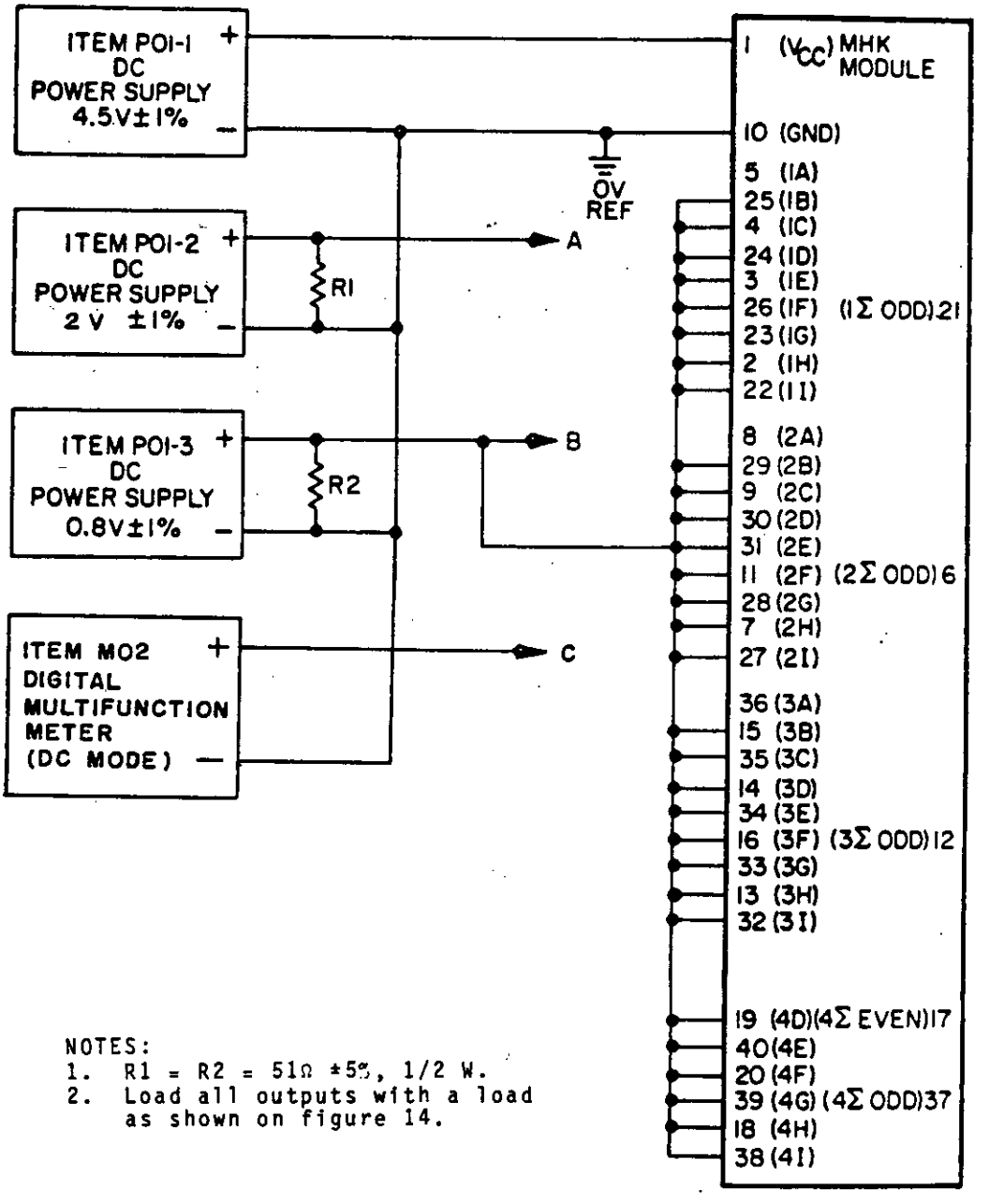
TABLE X. Output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1Σ odd Output voltage, low level	Open	5,8,36	21	0	500	mV	13
H2	2Σ odd "	"	"	6	"	"	"	"
H3	3Σ odd "	"	"	12	"	"	"	"
H4	4Σ odd "	"	"	37	"	"	"	"
H5	4Σ even "	36	5,8	17	"	"	"	"

TABLE XI. Output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	1Σ odd Output voltage, high level	5	8,36	21	2.5	4.5	V	13
J2	2Σ odd "	8	5,36	6	"	"	"	"
J3	3Σ odd "	36	5,8	12	"	"	"	"
J4	4Σ odd "	36	5,8	37	"	"	"	"
J5	4Σ even "	Open	5,8,36	17	"	"	"	"

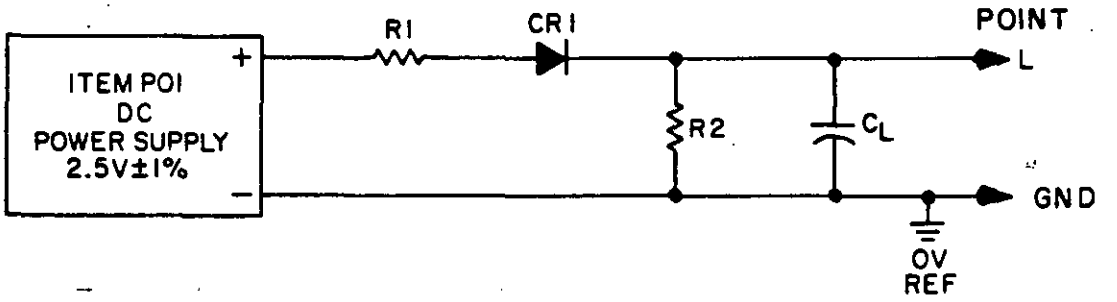
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs with a load as shown on figure 14.

* FIGURE 13. Output voltage (low-level and high-level) test circuit for tables X and XI.

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NOTES:
1. Load circuit 1 for 1Σ odd, 2Σ odd and 3Σ odd outputs:
   a. R1 = Select value for a current flow of 18 mA ±5% out of point L with point L held at 0.5 V.
   b. R2 = 2.61 kΩ ±1%, 1/8 W.
2. Load circuit 2 for 4Σ odd and 4Σ even outputs:
   a. R1 = Select value for a current flow of 20 mA ±5% out of point L with point L held at 0.5 V.
   b. R2 = 2.49 kΩ ±1%, 1/8 W.
3. Load circuits 1 and 2.
   a. CR1 = 1N3600, 1N4150, or equivalent.
   b. Test type S:
      C_L = 50 ±5 pF (including probe and parasitic capacitance).
      Test type Q:
      C_L = 150 ±5 pF (including probe and parasitic capacitance).
   c. Capacitor C_L is not required for test conditions H1-H5 and J1-J5.
   d. The lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm).

FIGURE 14. MHK logic load circuit.

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TABLE XII. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	1A Input current, high level	2,3,4,7,8,9,11,13,14,15, 16,18,19,20,22,23,24,25, 26,27,28,29,30,31,32,33, 34,35,36,38,39,40	5	0	50	µA	15
K2	1B "	2,3,4,5,7,8,9,11,13,14, 15,16,18,19,20,22,23,24, 26,27,28,29,30,31,32,33, 34,35,36,38,39,40	25	"	"	"	"
K3	1C "	2,3,5,7,8,9,11,13,14, 15,16,18,19,20,22,23,24, 25,26,27,28,29,30,31,32, 33,34,35,36,38,39,40	4	"	"	"	"
K4	1D "	2,3,4,5,7,8,9,11,13,14, 15,16,18,19,20,22,23, 25,26,27,28,29,30,31,32, 33,34,35,36,38,39,40	24	"	"	"	"
K5	1E "	2,4,5,7,8,9,11,13,14, 15,16,18,19,20,22,23,24, 25,26,27,28,29,30,31,32, 33,34,35,36,38,39,40	3	"	"	"	"
K6	1F "	2,3,4,5,7,8,9,11,13,14, 15,16,18,19,20,22,23,24, 25,27,28,29,30,31,32, 33,34,35,36,38,39,40	26	"	"	"	"
K7	1G "	2,3,4,5,7,8,9,11,13,14, 15,16,18,19,20,22,24, 25,26,27,28,29,30,31,32, 33,34,35,36,38,39,40	23	"	"	"	"
K8	1H "	3,4,5,7,8,9,11,13,14, 15,16,18,19,20,22,23,24, 25,26,27,28,29,30,31,32, 33,34,35,36,38,39,40	2	"	"	"	"
K9	1I "	2,3,4,5,7,8,9,11,13,14, 15,16,18,19,20,23,24, 25,26,27,28,29,30,31,32,	22	"	"	"	"

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TABLE XII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K10	2A Input current, high level	2,3,4,5,7,9,11,13,14, 15,16,18,19,20,22,23,24, 25,26,27,28,29,30,31,32, 33,34,35,36,38,39,40	8	0	50	µA	15
K11	2B "	2,3,4,5,7,8,9,11,13,14, 15,16,18,19,20,22,23,24, 25,26,27,28,30,31,32, 33,34,35,36,38,39,40	29	"	"	"	"
K12	2C "	2,3,4,5,7,8,11,13,14, 15,16,18,19,20,22,23,24, 25,26,27,28,29,30,31,32, 33,34,35,36,38,39,40	9	"	"	"	"
K13	2D "	2,3,4,5,7,8,9,11,13,14, 15,16,18,19,20,22,23,24, 25,26,27,28,29,31,32, 33,34,35,36,38,39,40	30	"	"	"	"
K14	2E "	2,3,4,5,7,8,9,11,13,14, 15,16,18,19,20,22,23,24, 25,26,27,28,29,30,32, 33,34,35,36,38,39,40	31	"	"	"	"
K15	2F "	2,3,4,5,7,8,9,13,14, 15,16,18,19,20,22,23,24, 25,26,27,28,29,30,31,32, 33,34,35,36,38,39,40	11	"	"	"	"
K16	2G "	2,3,4,5,7,8,9,11,13,14, 15,16,18,19,20,22,23,24, 25,26,27,29,30,31,32, 33,34,35,36,38,39,40	28	"	"	"	"
K17	2H "	2,3,4,5,8,9,11,13,14, 15,16,18,19,20,22,23,24, 25,26,27,28,29,30,31,32, 33,34,35,36,38,39,40	7	"	"	"	"
K18	2I "	2,3,4,5,7,8,9,11,13,14, 15,16,18,19,20,22,23,24, 25,26,28,29,30,31,32, 33,34,35,36,38,39,40	27	"	"	"	"

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TABLE XII. Input current (high-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K19	3A	Input current, high level	2,3,4,5,7,8,9,11,13,14, 15,16,18,19,20,22,23,24, 25,26,27,28,29,30,31,32, 33,34,35,38,39,40	36	0	50	µA	15
K20	3B	"	2,3,4,5,7,8,9,11,13,14, 16,18,19,20,22,23,24, 25,26,27,28,29,30,31,32, 33,34,35,36,38,39,40	15	"	"	"	"
K21	3C	"	2,3,4,5,7,8,9,11,13,14, 15,16,18,19,20,22,23,24, 25,26,27,28,29,30,31,32, 33,34,36,38,39,40	35	"	"	"	"
K22	3D	"	2,3,4,5,7,8,9,11,13, 15,16,18,19,20,22,23,24, 25,26,27,28,29,30,31,32, 33,34,35,36,38,39,40	14	"	"	"	"
K23	3E	"	2,3,4,5,7,8,9,11,13,14, 15,16,18,19,20,22,23,24, 25,26,27,28,29,30,31,32, 33,35,36,38,39,40	34	"	"	"	"
K24	3F	"	2,3,4,5,7,8,9,11,13,14, 15,18,19,20,22,23,24, 25,26,27,28,29,30,31,32, 33,34,35,36,38,39,40	16	"	"	"	"
K25	3G	"	2,3,4,5,7,8,9,11,13,14, 15,16,18,19,20,22,23,24, 25,26,27,28,29,30,31,32, 34,35,36,38,39,40	33	"	"	"	"
K26	3H	"	2,3,4,5,7,8,9,11,14, 15,16,18,19,20,22,23,24, 25,26,27,28,29,30,31,32, 33,34,35,36,38,39,40	13	"	"	"	"
K27	3I	"	2,3,4,5,7,8,9,11,13,14, 15,16,18,19,20,22,23,24, 25,26,27,28,29,30,31, 33,34,35,36,38,39,40	32	"	"	"	"

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TABLE XII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K28	4D Input current, high level	2,3,4,5,7,8,9,11,13,14, 15,16,18,20,22,23,24, 25,26,27,28,29,30,31,32, 33,34,35,36,38,39,40	19	0	50	µA	15
K29	4E "	2,3,4,5,7,8,9,11,13,14, 15,16,18,19,20,22,23,24, 25,26,27,28,29,30,31,32, 33,34,35,36,38,39,	40	"	"	"	"
K30	4F "	2,3,4,5,7,8,9,11,13,14, 15,16,18,19,22,23,24, 25,26,27,28,29,30,31,32, 33,34,35,36,38,39,40	20	"	"	"	"
K31	4G "	2,3,4,5,7,8,9,11,13,14, 15,16,18,19,20,22,23,24, 25,26,27,28,29,30,31,32, 33,34,35,36,38,40	39	"	"	"	"
K32	4H "	2,3,4,5,7,8,9,11,13,14, 15,16,19,20,22,23,24, 25,26,27,28,29,30,31,32, 33,34,35,36,38,39,40	18	"	"	"	"
K33	4I "	2,3,4,5,7,8,9,11,13,14, 15,16,18,19,20,22,23,24, 25,26,27,28,29,30,31,32, 33,34,35,36,39,40	38	"	"	"	"

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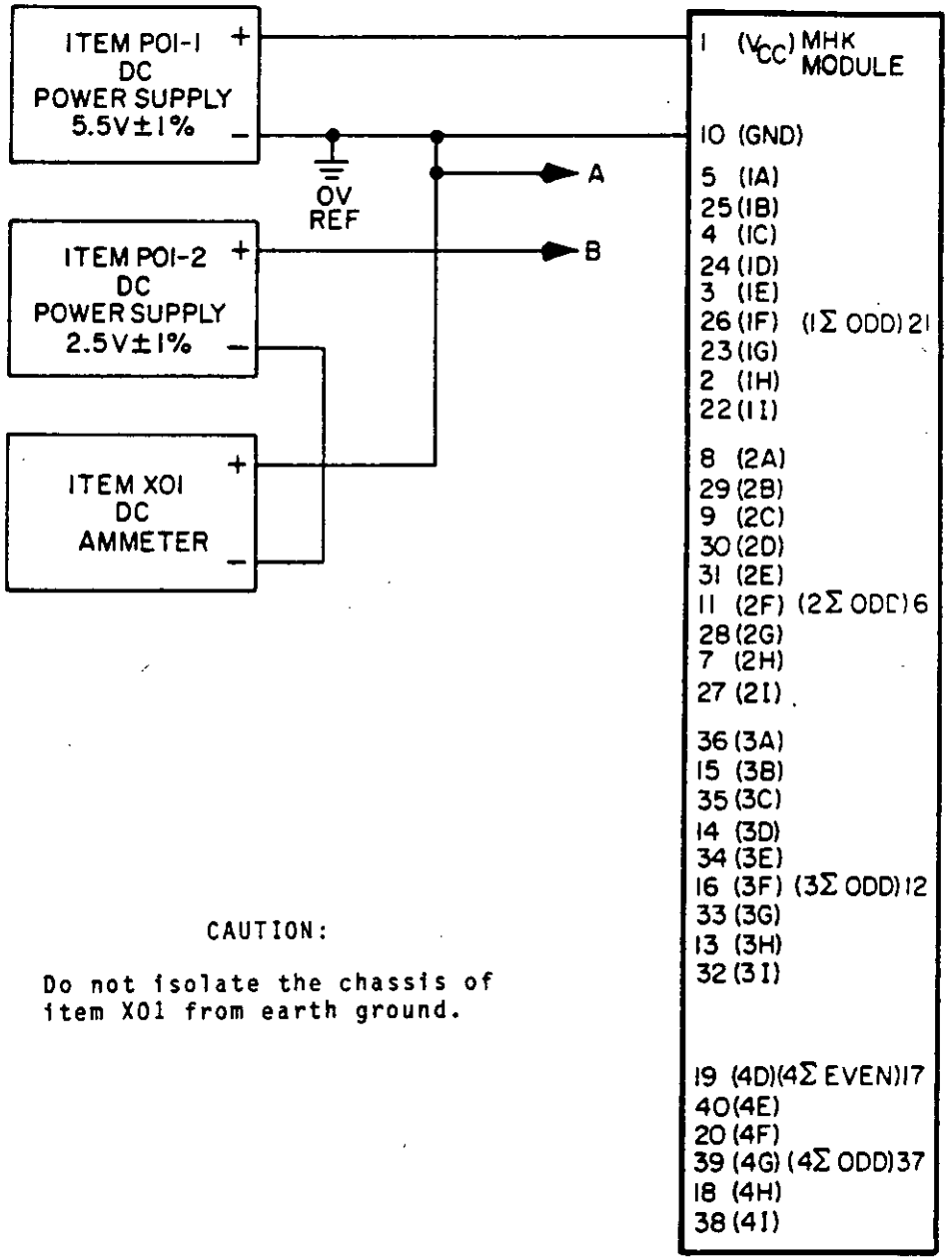


FIGURE 15. Input current (high-level) test circuit for table XII.

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TABLE XIII. Input current (low-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
L1	1A Input current, low level	5	-0.1	-2.2	mA	16
L2	1B "	25	"	"	"	"
L3	1C "	4	"	"	"	"
L4	1D "	24	"	"	"	"
L5	1E "	3	"	"	"	"
L6	1F "	26	"	"	"	"
L7	1G "	23	"	"	"	"
L8	1H "	2	"	"	"	"
L9	1I "	22	"	"	"	"
L10	2A "	8	"	"	"	"
L11	2B "	29	"	"	"	"
L12	2C "	9	"	"	"	"
L13	2D "	30	"	"	"	"
L14	2E "	31	"	"	"	"
L15	2F "	11	"	"	"	"
L16	2G "	28	"	"	"	"
L17	2H "	7	"	"	"	"
L18	2I "	27	"	"	"	"
L19	3A "	36	"	"	"	"
L20	3B "	15	"	"	"	"
L21	3C "	35	"	"	"	"

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TABLE XIII. Input current (low-level) test - Continued.

Test condition		Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
L22	3D	Input current, low level	14	-0.1	-2.2	mA	16
L23	3E	"	34	"	"	"	"
L24	3F	"	16	"	"	"	"
L25	3G	"	33	"	"	"	"
L26	3H	"	13	"	"	"	"
L27	3I	"	32	"	"	"	"
L28	4D	"	19	"	"	"	"
L29	4E	"	40	"	"	"	"
L30	4F	"	20	"	"	"	"
L31	4G	"	39	"	"	"	"
L32	4H	"	18	"	"	"	"
L33	4I	"	38	"	"	"	"

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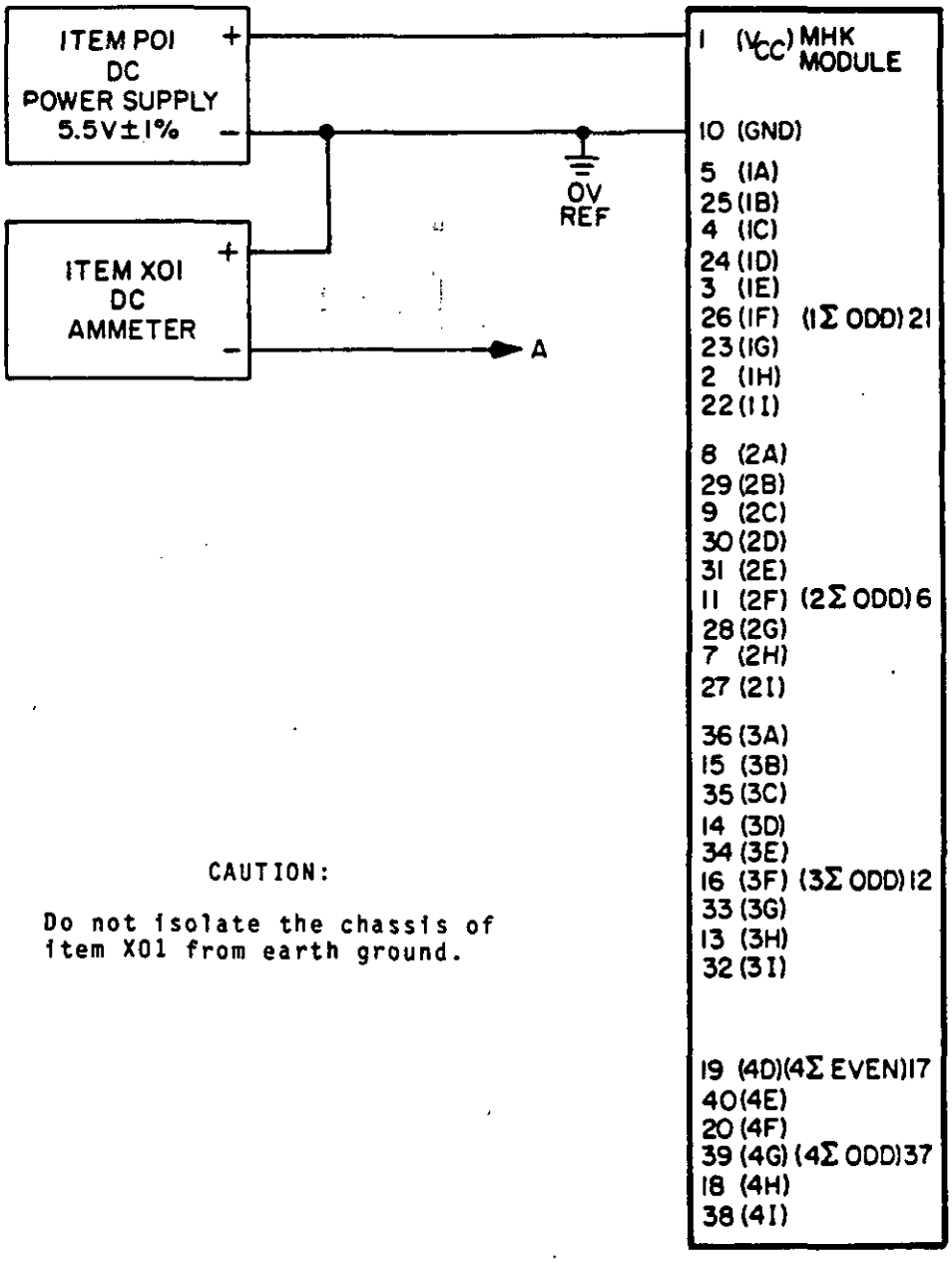


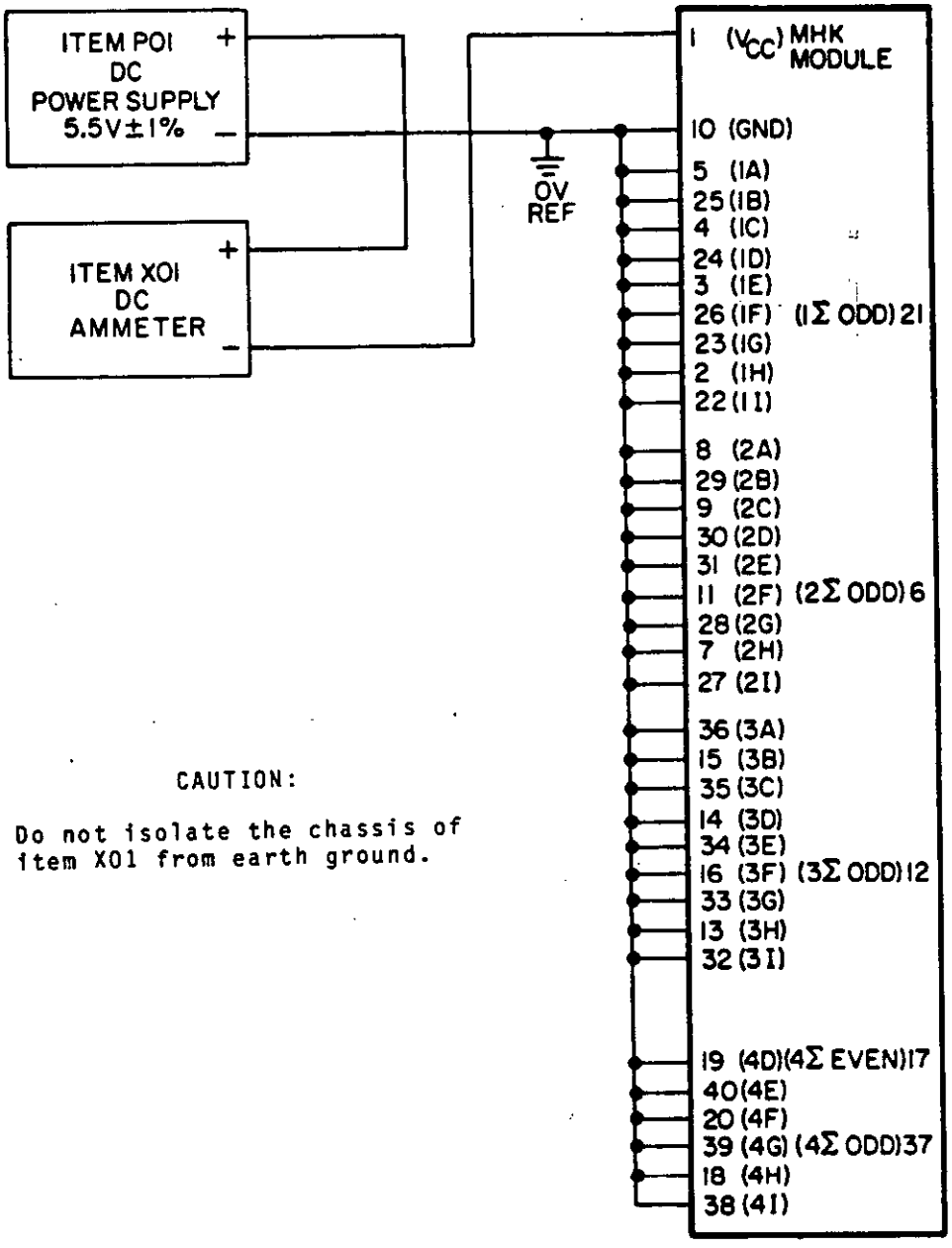
FIGURE 16. Input current (low-level) test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
M	Power supply current	100	396	mA	17

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

FIGURE 17. Power supply current test circuit for table XIV.

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TABLE XV. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	1Σ odd Transition time (t_{TLH}), C_L = 50 pF	5	21	Open	2,3,4,22,23,24, 25,26	2	15	ns	18,19
N2	2Σ odd "	8	6	"	7,9,11,27,28, 29,30,31	"	"	"	"
N3	3Σ odd "	36	12	"	13,14,15,16,32, 33,34,35	"	"	"	"
N4	4Σ odd "	19	37	"	5,8,18,20,36, 38,39,40	"	"	"	"
N5	4Σ even "	19	17	"	5,8,18,20,36, 38,39,40	"	"	"	"
N6	1Σ odd Transition time (t_{TLH}), C_L = 150 pF	5	21	"	2,3,4,22,23,24, 25,26	"	27	"	"
N7	2Σ odd "	8	6	"	7,9,11,27,28, 29,30,31	"	"	"	"
N8	3Σ odd "	36	12	"	13,14,15,16,32, 33,34,35	"	"	"	"
N9	4Σ odd "	19	37	"	5,8,18,20,36, 38,39,40	"	"	"	"
N10	4Σ even "	19	17	"	5,8,18,20,36, 38,39,40	"	"	"	"
P1	1Σ odd Transition time (t_{THL}), C_L = 50 pF	5	21	"	2,3,4,22,23,24, 25,26	"	6	"	"
P2	2Σ odd "	8	6	"	7,9,11,27,28, 29,30,31	"	"	"	"
P3	3Σ odd "	36	12	"	13,14,15,16,32, 33,34,35	"	"	"	"
P4	4Σ odd "	19	37	"	5,8,18,20,36, 38,39,40	"	"	"	"

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TABLE XV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P5	4Σ even Transition time (t_{THL}), C_L = 50 pF	19	17	Open	5,8,18,20,36, 38,39,40	2	6	ns	18,19
P6	1Σ odd Transition time (t_{THL}), C_L = 150 pF	5	21	"	2,3,4,22,23,24, 25,26	"	11	"	"
P7	2Σ odd "	8	6	"	7,9,11,27,28, 29,30,31	"	"	"	"
P8	3Σ odd "	36	12	"	13,14,15,16,32, 33,34,35	"	"	"	"
P9	4Σ odd "	19	37	"	5,8,18,20,36, 38,39,40	"	"	"	"
P10	4Σ even "	19	17	"	5,8,18,20,36, 38,39,40	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	1A to 1Σ odd — Propagation delay time (t_{PLH}), C_L = 50 pF	5	21	Open	2,22,23	5.0	27.5	ns	18,21
Q2	1B to 1Σ odd — "	25	"	"	3,4,5,24,26	"	"	"	"
Q3	1C to 1Σ odd — "	4	"	"	3,5,24,25,26	"	"	"	"
Q4	1C to 1Σ odd — "	4	"	"	2,3,22,23,24, 25,26	"	"	"	"
Q5	1D to 1Σ odd — "	24	"	"	4,5,25	"	"	"	"
Q6	2A to 2Σ odd — "	8	6	"	7,27,28	"	"	"	"
Q7	2B to 2Σ odd — "	29	"	"	8,9,11,30,31	"	"	"	"
Q8	2C to 2Σ odd — "	9	"	"	8,11,29,30,31	"	"	"	"
Q9	2C to 2Σ odd — "	9	"	"	7,11,27,28,29, 30,31	"	"	"	"
Q10	2D to 2Σ odd — "	30	"	"	8,9,29	"	"	"	"
Q11	3A to 3Σ odd — "	36	12	"	13,32,33	"	"	"	"
Q12	3B to 3Σ odd — "	15	"	"	14,16,34,35,36	"	"	"	"
Q13	3C to 3Σ odd — "	35	"	"	14,15,16,34,36	"	"	"	"
Q14	3C to 3Σ odd — "	35	"	"	13,14,15,16,32, 33,34	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q15	3D to 3Σ odd; Propagation delay time (t_{PLH}), C_L = 50 pF	14	12	Open	15,35,36	5.0	27.5	ns	18,21
Q16	4D to 4Σ odd; "	19	37	"	5,8,18,36,38, 39,40	"	"	"	"
Q17	4E to 4Σ odd; "	40	"	"	18,19,20,38,39	"	"	"	"
Q18	4F to 4Σ odd; "	20	"	"	18,38,39	"	"	"	"
Q19	4D to 4Σ even; "	19	17	"	5,8,18,20,36, 38,39,40	"	"	"	"
Q20	4E to 4Σ even; "	40	"	"	5,8,18,19,20, 36,38,39	"	"	"	"
Q21	4F to 4Σ even; "	20	"	"	19,40	"	"	"	"
Q22	1D to 4Σ odd; "	24	37	21	2,3,8,19,20,22, 23,26,36,40	5	55	"	18,20
Q23	2D to 4Σ odd; "	30	"	6	5,7,11,19,20, 27,28,31,36,40	"	"	"	"
Q24	3D to 4Σ odd; "	14	"	12	5,8,13,16,19, 20,32,33,34,40	"	"	"	"
Q25	1G to 4Σ even; "	23	17	21	2,3,8,18,19,20, 22,24,26,36,38, 39,40	"	"	"	"
Q26	2G to 4Σ even; "	28	"	6	5,7,11,18,19, 20,27,30,31,36, 38,39,40	"	"	"	"
Q27	3G to 4Σ even; "	33	"	12	13,14,16,18,19, 20,32,34,38,39, 40	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q28	1A to 1Σ odd — Propagation delay time (t_{PLH}), C_L = 150 pF	5	21	Open	2,22,23	5	32	ns	18,21
Q29	1B to 1Σ odd — "	25	"	"	3,4,5,24,26	"	"	"	"
Q30	1C to 1Σ odd — "	4	"	"	3,5,24,25,26	"	"	"	"
Q31	1C to 1Σ odd — "	4	"	"	2,3,22,23,24, 25,26	"	"	"	"
Q32	1D to 1Σ odd — "	24	"	"	4,5,25	"	"	"	"
Q33	2A to 2Σ odd — "	8	6	"	7,27,28	"	"	"	"
Q34	2B to 2Σ odd — "	29	"	"	8,9,11,30,31	"	"	"	"
Q35	2C to 2Σ odd — "	9	"	"	8,11,29,30,31	"	"	"	"
Q36	2C to 2Σ odd — "	9	"	"	7,11,27,28,29, 30,31	"	"	"	"
Q37	2D to 2Σ odd — "	30	"	"	8,9,29	"	"	"	"
Q38	3A to 3Σ odd — "	36	12	"	13,32,33	"	"	"	"
Q39	3B to 3Σ odd — "	15	"	"	14,16,34,35,36	"	"	"	"
Q40	3C to 3Σ odd — "	35	"	"	14,15,16,34,36	"	"	"	"
Q41	3C to 3Σ odd — "	35	"	"	13,14,15,16,32, 33,34	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q42	3D to 3Σ odd	Propagation delay time (t_{PLH}), C_L = 150 pF	14	12	Open	15,35,36	5	32	ns	18,21
Q43	4D to 4Σ odd	"	19	37	"	5,8,18,36,38, 39,40	"	"	"	"
Q44	4E to 4Σ odd	"	40	"	"	18,19,20,38,39	"	"	"	"
Q45	4F to 4Σ odd	"	20	"	"	18,38,39	"	"	"	"
Q46	4D to 4Σ even	"	19	17	"	5,8,18,20,36, 38,39,40	"	"	"	"
Q47	4E to 4Σ even	"	40	"	"	5,8,18,19,20, 36,38,39	"	"	"	"
Q48	4F to 4Σ even	"	20	"	"	19,40	"	"	"	"
Q49	1D to 4Σ odd	"	24	37	21	2,3,8,19,20,22, 23,26,36,40	"	64	"	18,20
Q50	2D to 4Σ odd	"	30	"	6	5,7,11,19,20, 27,28,31,36,40	"	"	"	"
Q51	3D to 4Σ odd	"	14	"	12	5,8,13,16,19, 20,32,33,34,40	"	"	"	"
Q52	1G to 4Σ even	"	23	17	21	2,3,8,18,19,20, 22,24,26,36,38, 39,40	"	"	"	"
Q53	2G to 4Σ even	"	28	"	6	5,7,11,18,19, 20,27,30,31,36, 38,39,40	"	"	"	"
Q54	3G to 4Σ even	"	33	"	12	13,14,16,18,19, 20,32,34,38,39, 40	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1E to 1Σ odd	Propagation delay time (t_{PHL}), C_L = 50 pF	3	21	Open	4,5,24,25,26	5	24	ns	18,21
R2	1F to 1Σ odd	"	26	"	"	3,4,5,24,25	"	"	"	"
R3	1G to 1Σ odd	"	23	"	"	2,3,4,5,24,25, 26	"	"	"	"
R4	1H to 1Σ odd	"	2	"	"	3,22,23,24,26	"	"	"	"
R5	1I to 1Σ odd	"	22	"	"	3,24,26	"	"	"	"
R6	2E to 2Σ odd	"	31	6	"	8,9,11,29,30	"	"	"	"
R7	2F to 2Σ odd	"	11	"	"	8,9,29,30,31	"	"	"	"
R8	2G to 2Σ odd	"	28	"	"	7,8,9,11,29, 30,31	"	"	"	"
R9	2H to 2Σ odd	"	7	"	"	11,27,28,30,31	"	"	"	"
R10	2I to 2Σ odd	"	27	"	"	11,30,31	"	"	"	"
R11	3E to 3Σ odd	"	34	12	"	14,15,16,35,36	"	"	"	"
R12	3F to 3Σ odd	"	16	"	"	14,15,34,35,36	"	"	"	"
R13	3G to 3Σ odd	"	33	"	"	13,14,15,16,34, 35,36	"	"	"	"
R14	3H to 3Σ odd	"	13	"	"	14,16,32,33,34	"	"	"	"
R15	3I to 3Σ odd	"	32	"	"	14,16,34	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R16	4D to 4Σ odd	Propagation delay time (t_{PHL}), C_L = 50 pF	19	37	Open	5,8,18,36,38, 39,40	5	24	ns	18,21
R17	4G to 4Σ odd	"	39	"	"	5,8,36	"	"	"	"
R18	4H to 4Σ odd	"	18	"	"	5,8,36,38,39	"	"	"	"
R19	4I to 4Σ odd	"	38	"	"	5,8,18,36,39	"	"	"	"
R20	4D to 4Σ even	"	19	17	"	5,8,20,36	"	"	"	"
R21	4G to 4Σ even	"	39	"	"	5,8,18,19,20, 36,38,40	"	"	"	"
R22	4H to 4Σ even	"	18	"	"	5,8,19,20,36, 38,39,40	"	"	"	"
R23	4I to 4Σ even	"	38	"	"	5,8,18,19,20, 36,39,40	"	"	"	"
R24	1D to 4Σ odd	"	24	37	21	2,3,8,19,20,22, 23,26,36,40	5.0	51.5	"	18,20
R25	2D to 4Σ odd	"	30	"	6	5,7,11,19,20, 27,28,31,36,40	"	"	"	"
R26	3D to 4Σ odd	"	14	"	12	5,8,13,16,19, 20,32,33,34,40	"	"	"	"
R27	1G to 4Σ even	"	23	17	21	2,3,8,18,19,20, 22,24,26,36,38, 39,40	"	"	"	"
R28	2G to 4Σ even	"	28	"	6	5,7,11,18,19, 20,27,30,31,36, 38,39,40	"	"	"	"
R29	3G to 4Σ even	"	33	"	12	13,14,16,18,19, 20,32,34,38,39, 40	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R30	1E to 1Σ odd	Propagation delay time (t_{PHL}), C_L = 150 pF	3	21	Open	4,5,24,25,26	5	28	ns	18,21
R31	1F to 1Σ odd	"	26	"	"	3,4,5,24,25	"	"	"	"
R32	1G to 1Σ odd	"	23	"	"	2,3,4,5,24,25, 26	"	"	"	"
R33	1H to 1Σ odd	"	2	"	"	3,22,23,24,26	"	"	"	"
R34	1I to 1Σ odd	"	22	"	"	3,24,26	"	"	"	"
R35	2E to 2Σ odd	"	31	6	"	8,9,11,29,30	"	"	"	"
R36	2F to 2Σ odd	"	11	"	"	8,9,29,30,31	"	"	"	"
R37	2G to 2Σ odd	"	28	"	"	7,8,9,11,29,30, 31	"	"	"	"
R38	2H to 2Σ odd	"	7	"	"	11,27,28,30,31	"	"	"	"
R39	2I to 2Σ odd	"	27	"	"	11,30,31	"	"	"	"
R40	3E to 3Σ odd	"	34	12	"	14,15,16,35,36	"	"	"	"
R41	3F to 3Σ odd	"	16	"	"	14,15,34,35,36	"	"	"	"
R42	3G to 3Σ odd	"	33	"	"	13,14,15,16,34, 35,36	"	"	"	"
R43	3H to 3Σ odd	"	13	"	"	14,16,32,33,34	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

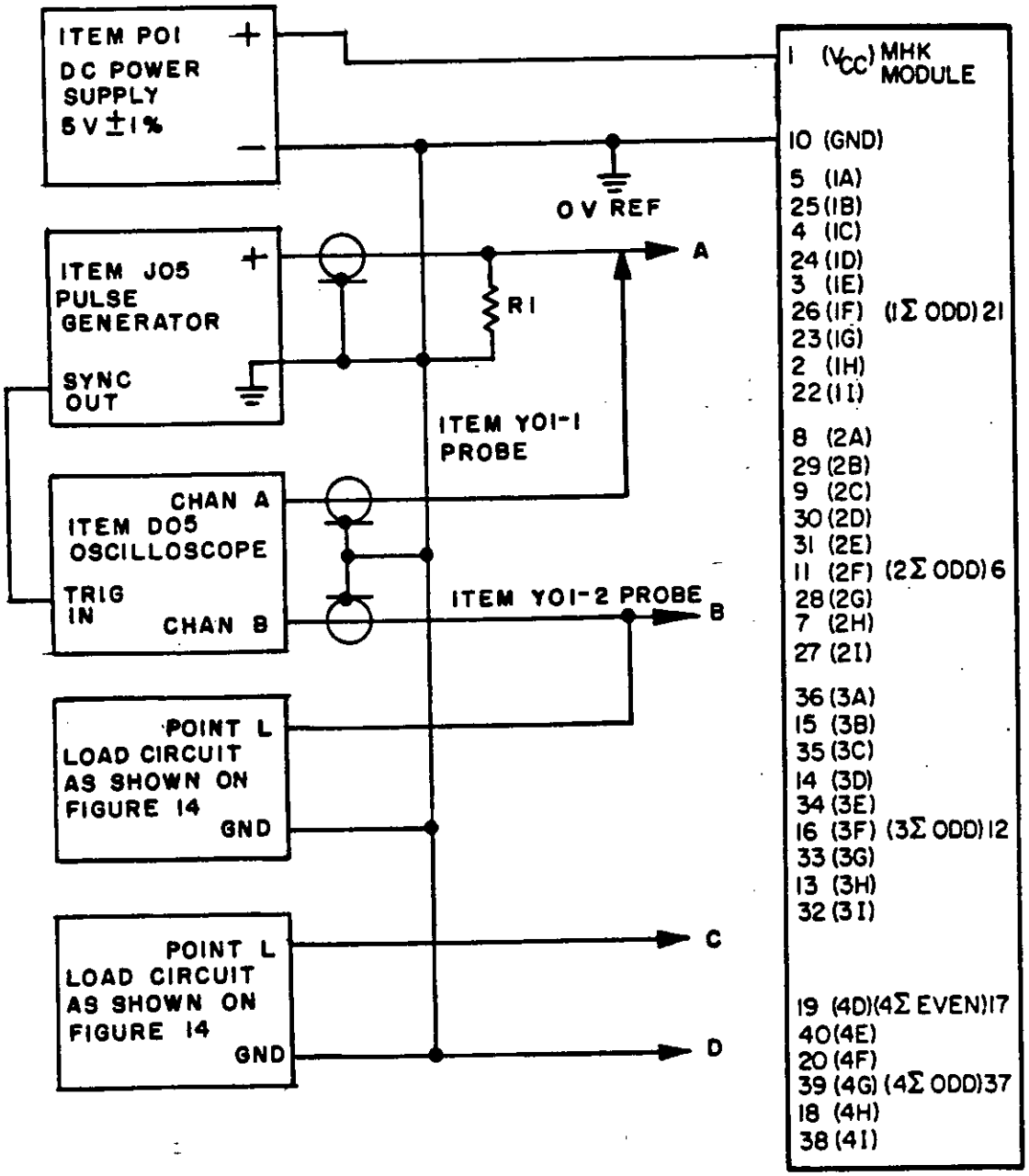
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R44	3I to 3Σ odd	Propagation delay time (t_{PHL}), C_L = 150 pF	32	12	Open	14,16,34	5	28	ns	18,21
R45	4D to 3Σ odd	"	19	37	"	5,8,18,36,38, 39,40	"	"	"	"
R46	4G to 4Σ odd	"	39	"	"	5,8,36	"	"	"	"
R47	4H to 4Σ odd	"	18	"	"	5,8,36,38,39	"	"	"	"
R48	4I to 4Σ odd	"	38	"	"	5,8,18,36,39	"	"	"	"
R49	4D to 4Σ even	"	19	17	"	5,8,20,36	"	"	"	"
R50	4G to 4Σ even	"	39	"	"	5,8,18,19,20, 36,38,40	"	"	"	"
R51	4H to 4Σ even	"	18	"	"	5,8,19,20,36, 38,39,40	"	"	"	"
R52	4I to 4Σ even	"	38	"	"	5,8,18,19,20, 36,39,40	"	"	"	"
R53	1D to 4Σ odd	"	24	37	21	2,3,8,19,20,22, 23,26,36,40	"	60	"	18,20
R54	2D to 4Σ odd	"	30	"	6	5,7,11,19,20, 27,28,31,36,40	"	"	"	"
R55	3D to 4Σ odd	"	14	"	12	5,8,13,16,19, 20,32,33,34,40	"	"	"	"
R56	1G to 4Σ even	"	23	17	21	2,3,8,18,19,20, 22,24,26,36,38, 39,40	"	"	"	"

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R57	2G to 4Σ even Propagation delay time (t_{PHL}), C_L = 150 pF	28	17	6	5,7,11,18,19, 20,27,30,31,38, 39,36,40	5	60	ns	18,20
R58	3G to 4Σ even Propagation delay time (t_{PHL}), C_L = 150 pF	33	17	12	13,14,16,18,19, 20,32,34,38,39, 40	5	60	ns	18,20

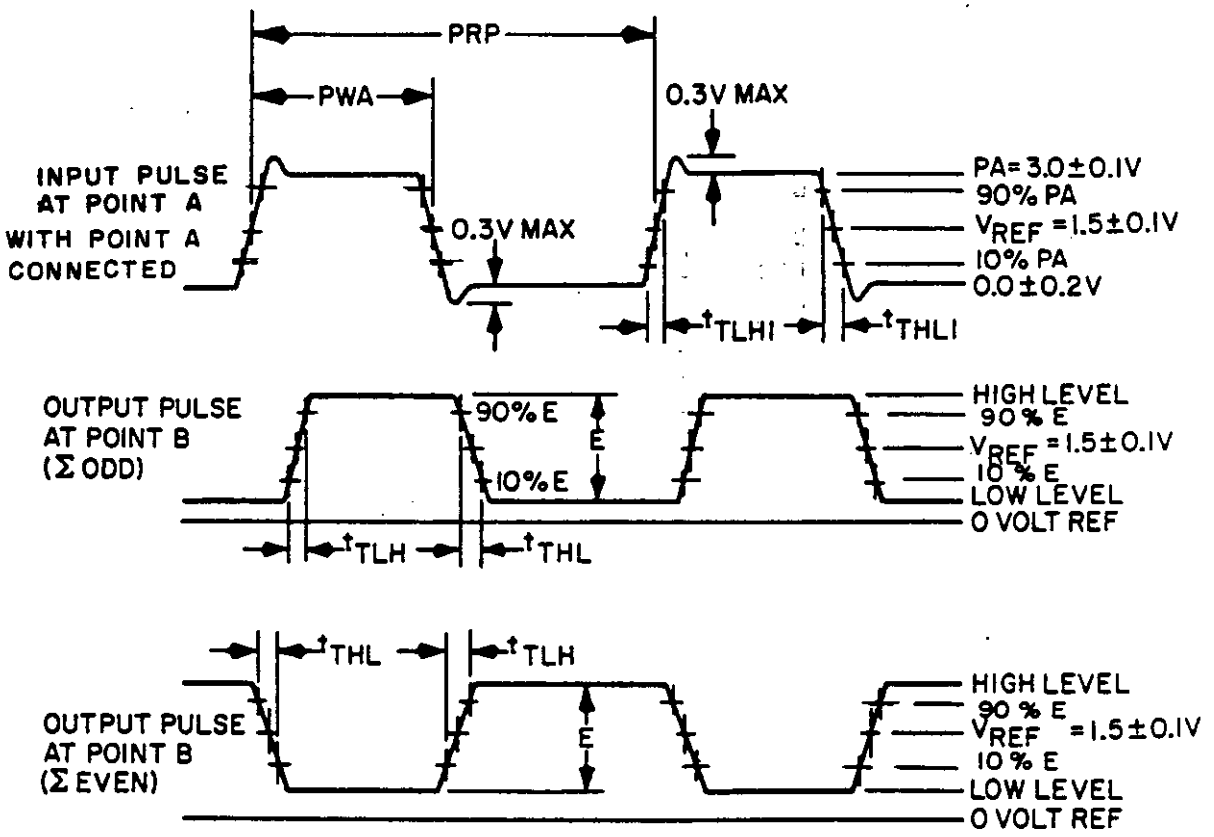
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load all outputs with a load as shown on figure 14.

* FIGURE 18. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XV and XVI.

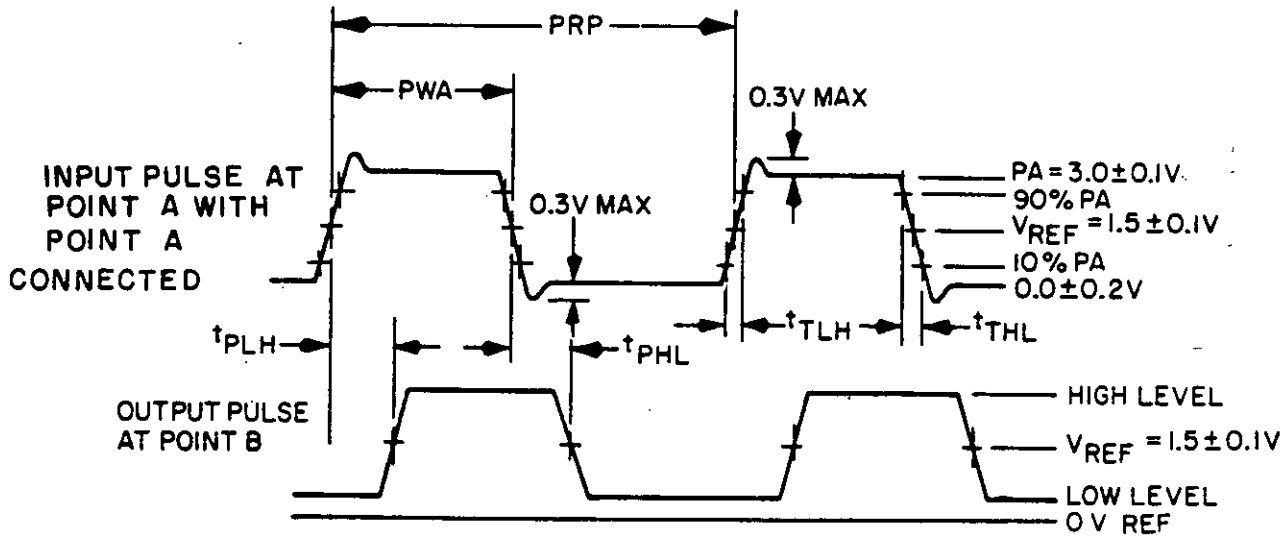
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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 4 \pm 1$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between resistor R1 and input pins; between items YO1-1 and YO1-2 and output pins; and between load circuits and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 19. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for table XV.

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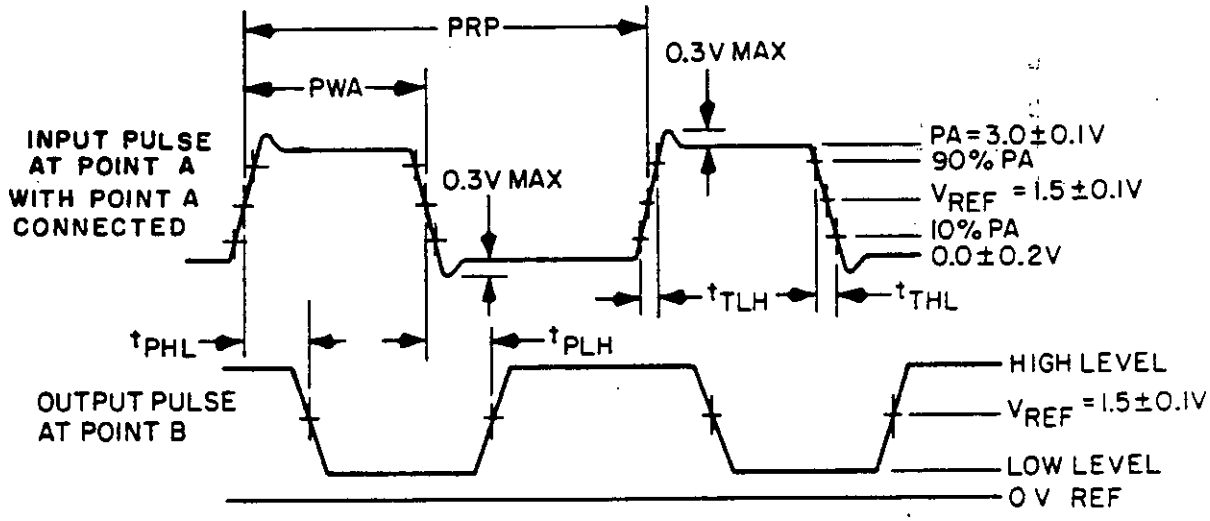


NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.

   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between resistor R1 and input pins; between items Y01-1 and Y01-2 and output pins; and between load circuits and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 20. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XVI.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.

   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements, verify that the output pulse high level i
   ≥ 2.5 V and the low level is ≤ 0.5 V.
3. The lead length of electrical conductors between resistor R1 and input pi
   between items Y01-1 and Y01-2 and output pins; and between load circuits
   output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

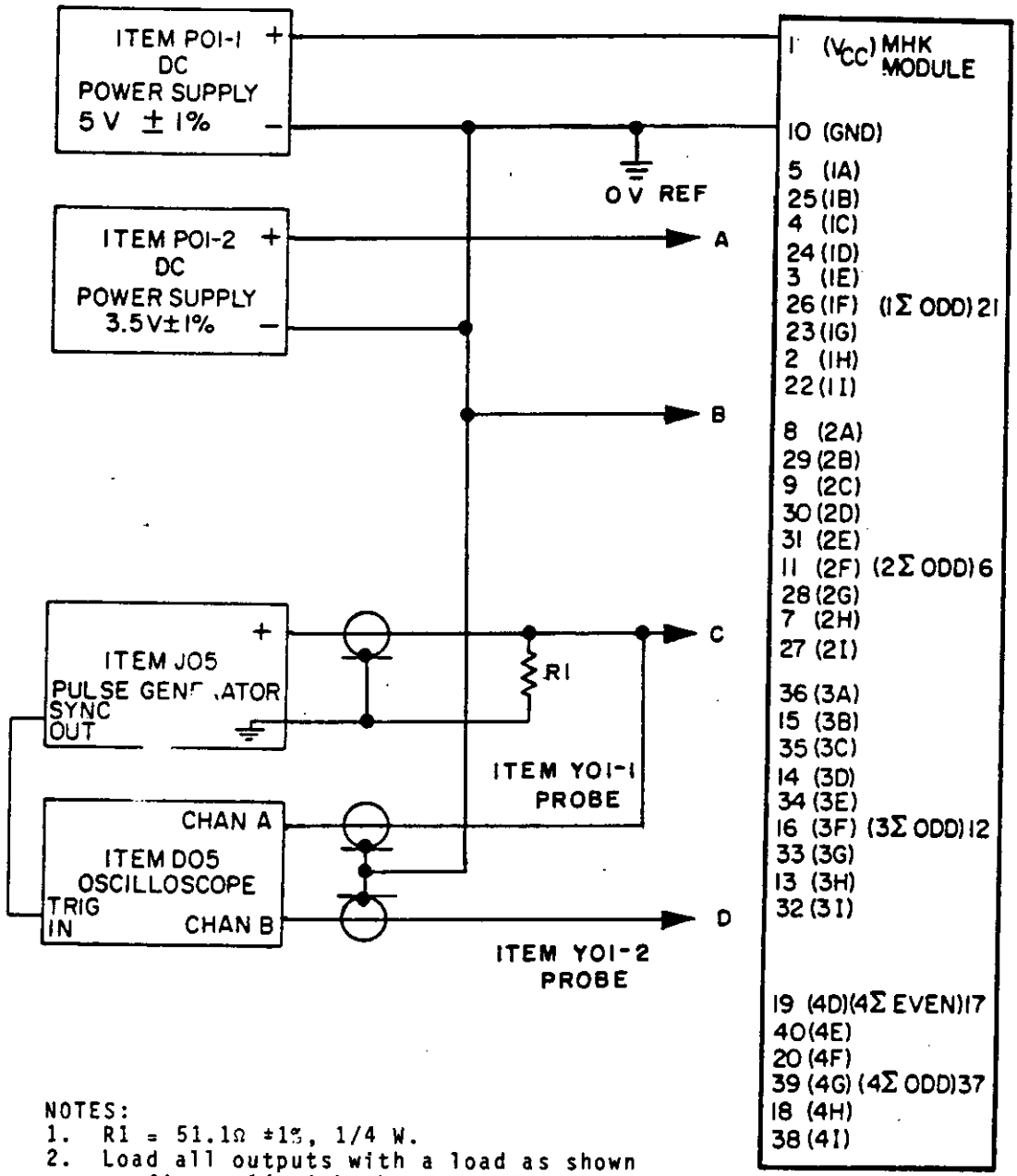
FIGURE 21. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XVI.

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TABLE XVII. Output crosstalk (low-level and high-level) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1Σ odd	Output crosstalk, low level	2,3,4,7,9,11,13,14,15,16,18, 20,22,23,24,25,26,27,28,29, 30,31,32,33,34,35,38,39,40	5	8,19,36	21	0	500	mV	22,23
S2	2Σ odd	"	2,3,4,7,9,11,13,14,15,16,18, 20,22,23,24,25,26,27,28,29, 30,31,32,33,34,35,38,39,40	8	5,19,36	6	"	"	"	"
S3	3Σ odd	"	2,3,4,7,9,11,13,14,15,16,18, 20,22,23,24,25,26,27,28,29, 30,31,32,33,34,35,38,39,40	36	5,18,19	12	"	"	"	"
T1	1Σ odd	Output crosstalk, high level	5,7,9,11,13,14,15,16,18,20, 27,28,29,30,31,32,33,34,35, 38,39,40	2,3,4,22, 23,24,25, 26	8,19,36	21	"	400	"	"
T2	2Σ odd	"	2,3,4,8,13,14,15,16,18,20, 22,23,24,25,26,32,33,34,35, 38,39,40	7,9,11,27, 28,29,30, 31	5,19,36	6	"	"	"	"
T3	3Σ odd	"	2,3,4,7,9,11,18,20,22,23,24, 25,26,27,28,29,30,31,36,38, 39,40	13,14,15, 16,32,33, 34,35	5,8,19	12	"	"	"	"

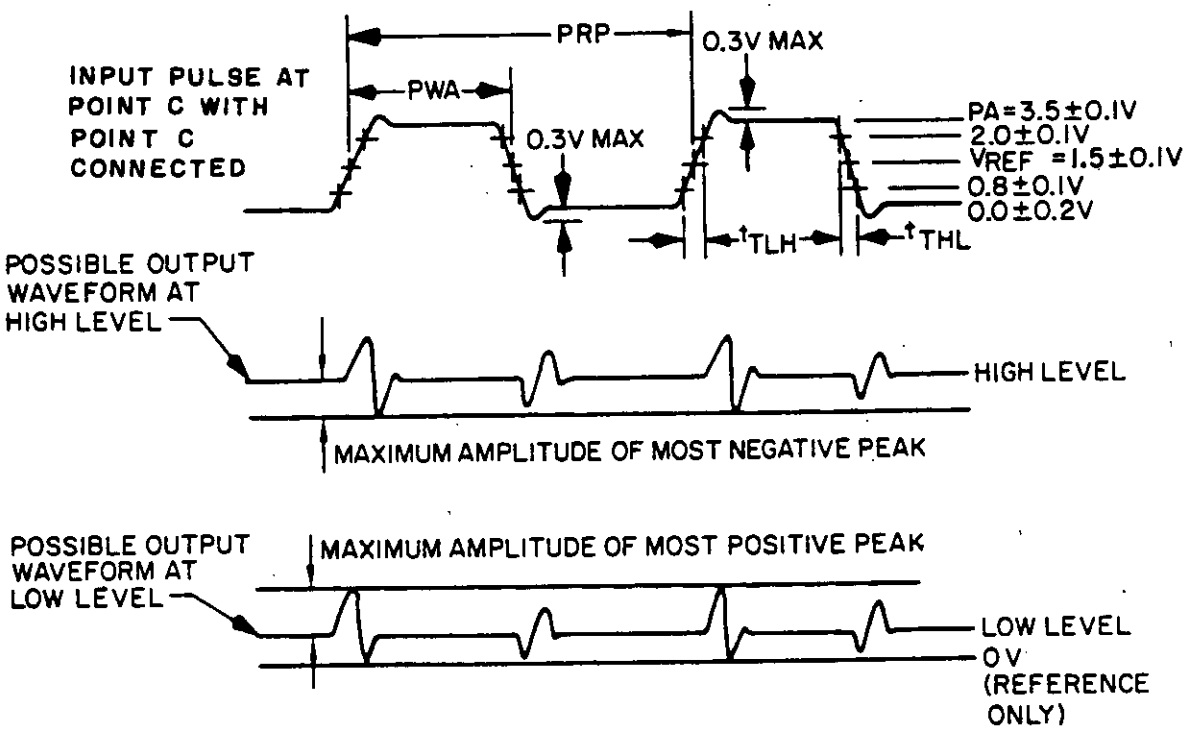
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load all outputs with a load as shown on figure 14 with the exception that C_L = 50 ±5 pF.

FIGURE 22. Output crosstalk (low-level and high-level) test circuit for table XVII.

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NOTE: Input pulse parameters:
PRP = 1,000 ±100 ns.
PWA = 500 ±50 ns.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).

FIGURE 23. Output crosstalk (low-level and high-level) test circuit waveforms for table XVII.

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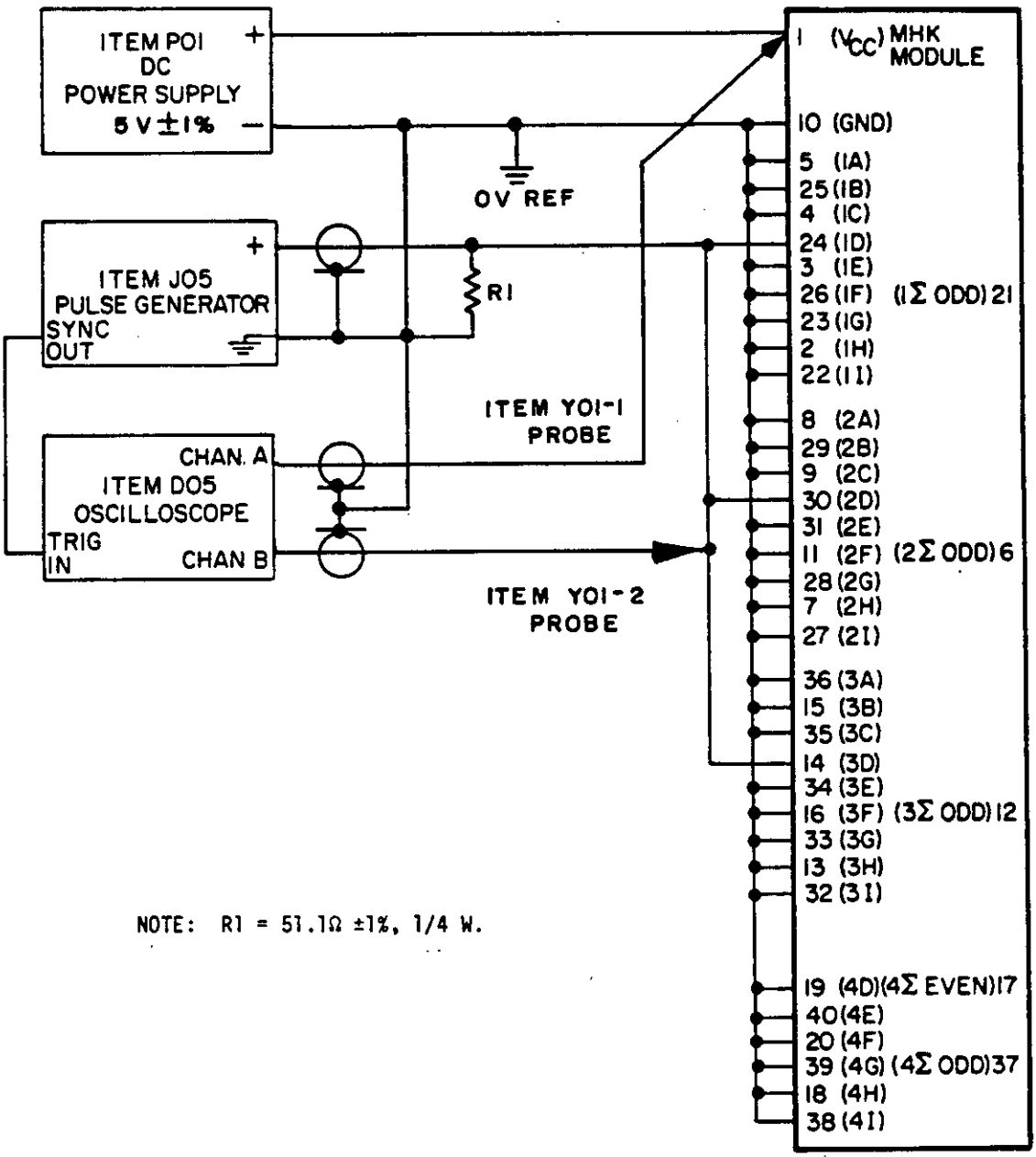
TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
U	Power supply transient amplitude	0	400	mV	24,25

TABLE XIX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V	Power supply filter capacitance	Open	1	1.6	3.0	µF	26

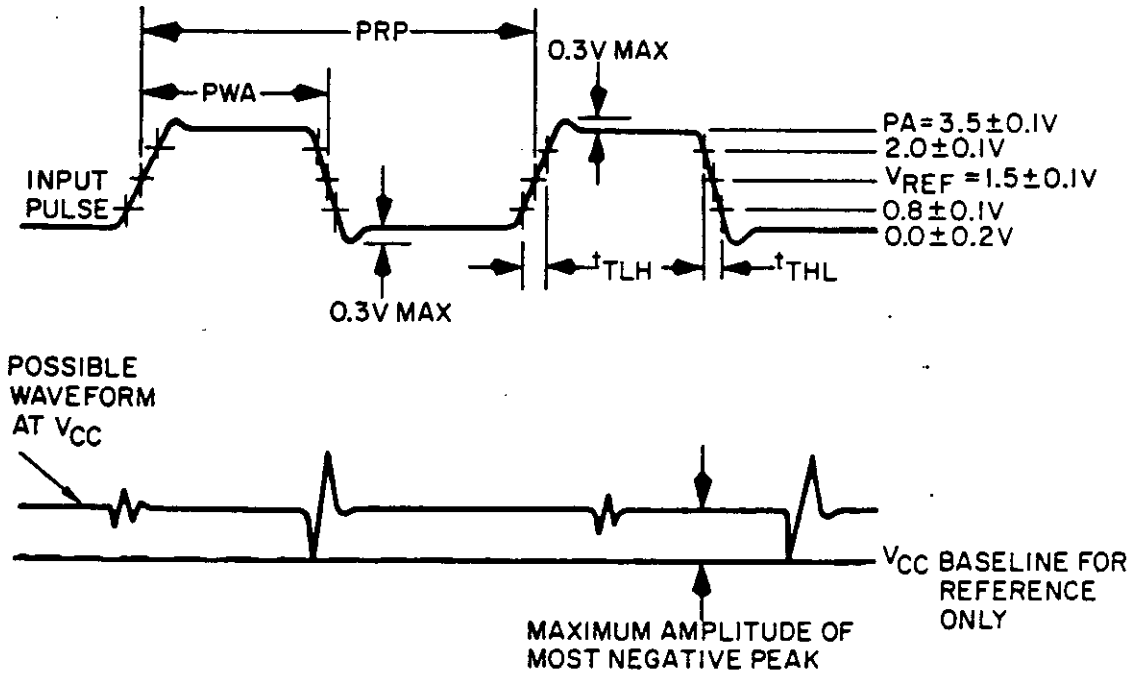
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NOTE: R1 = 51.1Ω ±1%, 1/4 W.

FIGURE 24. Power supply transient amplitude test circuit for table XVIII.

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NOTE: Input pulse parameters:
PRP = 1,000 ±100 ns.
PWA = 500 ±50 ns.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).

FIGURE 25. Power supply transient amplitude test circuit waveforms for table XVIII.

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TABLE XX. Input capacitance test.

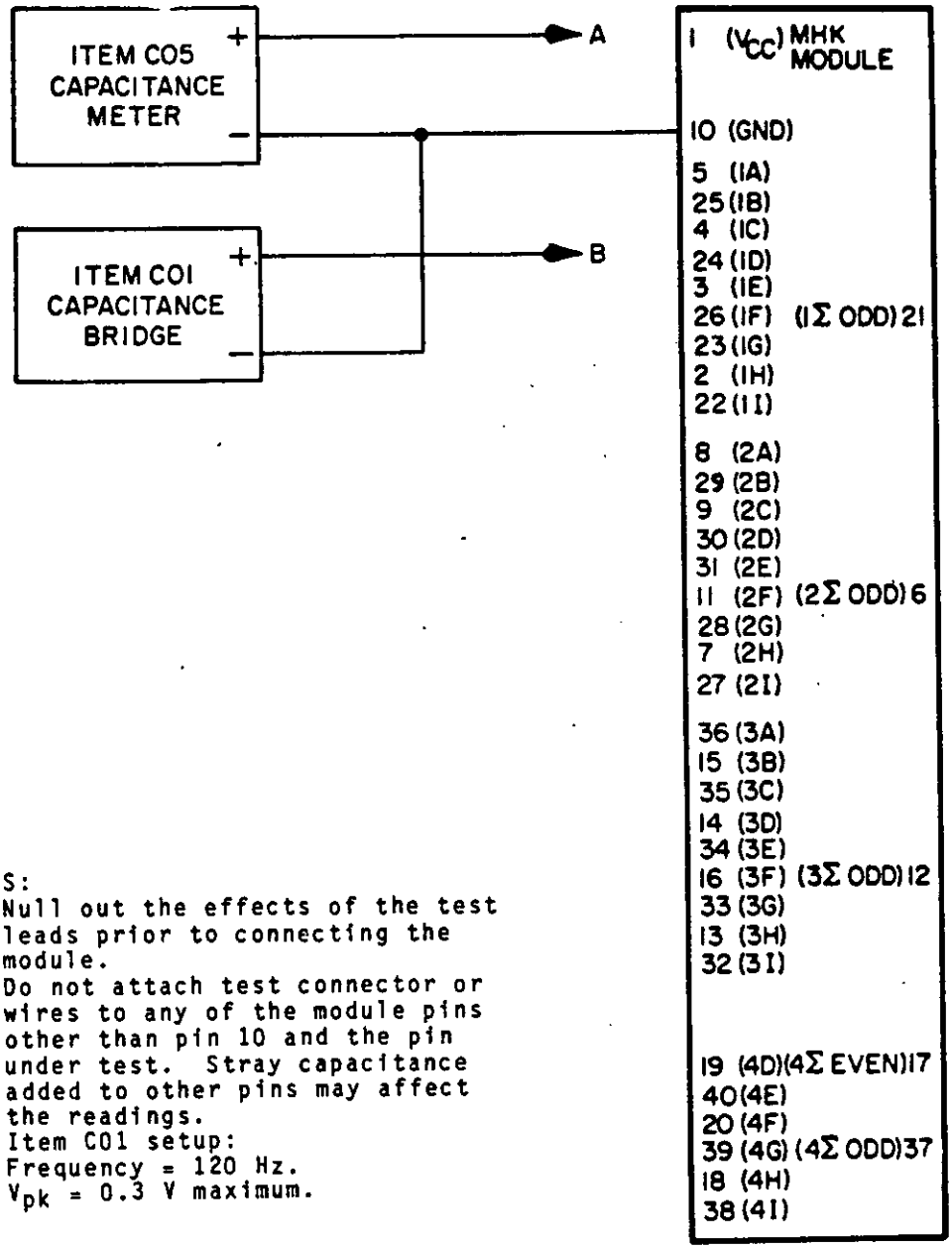
Test condition		Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
W1	1A	Input capacitance	5	Open	5	15	pF	26
W2	1B	"	25	"	"	"	"	"
W3	1C	"	4	"	"	"	"	"
W4	1D	"	24	"	"	"	"	"
W5	1E	"	3	"	"	"	"	"
W6	1F	"	26	"	"	"	"	"
W7	1G	"	23	"	"	"	"	"
W8	1H	"	2	"	"	"	"	"
W9	1I	"	22	"	"	"	"	"
W10	2A	"	8	"	"	"	"	"
W11	2B	"	29	"	"	"	"	"
W12	2C	"	9	"	"	"	"	"
W13	2D	"	30	"	"	"	"	"
W14	2E	"	31	"	"	"	"	"
W15	2F	"	11	"	"	"	"	"
W16	2G	"	28	"	"	"	"	"
W17	2H	"	7	"	"	"	"	"
W18	2I	"	27	"	"	"	"	"
W19	3A	"	36	"	"	"	"	"
W20	3B	"	15	"	"	"	"	"
W21	3C	"	35	"	"	"	"	"

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TABLE XX. Input capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W22	3D Input capacitance	14	Open	5	15	pF	26
W23	3E "	34	"	"	"	"	"
W24	3F "	16	"	"	"	"	"
W25	3G "	33	"	"	"	"	"
W26	3H "	13	"	"	"	"	"
W27	3I "	32	"	"	"	"	"
W28	4D "	19	"	"	"	"	"
W29	4E "	40	"	"	"	"	"
W30	4F "	20	"	"	"	"	"
W31	4G "	39	"	"	"	"	"
W32	4H "	18	"	"	"	"	"
W33	4I "	38	"	"	"	"	"

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

FIGURE 26. Power supply filter and input capacitance test circuit for tables XIX and XX.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/77A(EC)	Modules, Standard Electronic Encoder, Parity, Digital, Key Code MHK

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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